

A PROJECT ON
“Institute Management System (IMS)”

SUBMITTED IN
PARTIAL FULFILLMENT OF THE REQUIREMENT
FOR THE COURSE OF
DIPLOMA IN ADVANCED COMPUTING FROM CDAC



SUNBEAM INSTITUTE OF INFORMATION TECHNOLOGY
Hinjawadi

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UNDER THE GUIDANCE OF:

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ACKNOWLEDGEMENT

A project usually falls short of its expectation unless aided and guided by the right persons at the right time. We avail this opportunity to express our deep sense of gratitude towards Mr. Nitin Kudale (Center Coordinator, SIIT, Pune) and Mr. Yogesh Kolhe (Course Coordinator, SIIT ,Pune) .

We are deeply indebted and grateful to them for their guidance, encouragement and deep concern for our project. Without their critical evaluation and suggestions at every stage of the project, this project could never have reached its present form.

Last but not the least we thank the entire faculty and the staff members of Sunbeam Institute of Information Technology, Pune for their support.

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CERTIFICATE

This is to certify that the project work under the title 'Web Portal for Student and teacher' is done by Pimpalkar Prathamesh Dilip, Teerth Budhaliya, Dheeraj Ghosh And Pachpande Harshal Rajendra in partial fulfillment of the requirement for award of Diploma in Advanced Computing Course.

Mrs. Pooja Jaiswal
Project Guide
Date: 03/02/2026

Mr. Yogesh Kolhe
Course Co-Coordinator

1. INTRODUCTION TO PROJECT

The web-based "Institute Management System" project is an attempt to automate and streamline the core operations of an educational institute. The system enables different stakeholders — Admin, Teacher, and Student — to manage academic and administrative activities efficiently through a unified digital platform.

The system provides role-based access:

Admin can manage students, teachers, courses, attendance, marks, notices, subjects, and feedbacks.

Teacher can view student details, attendance, marks, courses, schedule, institute notices, evaluate students, Schedule exam.

Students can access their profile, view attendance and give complaints, and check notices and give scheduled exams.

The system simplifies tasks such as adding new students or teachers, recording attendance, managing marks, and publishing notices. It ensures real-time data availability so that all stakeholders can make informed decisions.

The 'Admin' dashboard allows quick access to all institute data and operations without the need for manual record-keeping. Teachers can directly update or check academic records, and Schedule exams while students have a self-service portal to track their academic progress and institute updates.

For example, the Student Module allows students to log in and view personal details, attendance records, marks, institute notices, and appear for exams. Similarly, the Admin Module enables quick student enrollment, teacher assignment, subject allocation, and feedback resolution.

Overall, the system reduces paperwork, improves efficiency, and keeps centralized, secure records for smooth institute management operations.

2.REQUIREMENTS

2.1 FUNCTIONAL REQUIREMENTS



2.1 Admin Functionality

1. Student Administration

The Admin plays a central role in managing the student lifecycle within the Institute Management System. Using the Add, Edit, Delete a Student functionality, the admin can maintain an accurate and up-to-date database of all enrolled students. This ensures that new students are onboarded correctly and records are modified or removed as necessary. To keep track of the entire student body, the admin utilizes the

Display All Students feature, which provides a comprehensive list of all active students for easy reference.

2. Teacher Administration

The system empowers the admin to oversee faculty members effectively. Through the Add, Edit, Delete a Teacher use case, the admin can manage faculty profiles, ensuring that only authorized teaching staff have access to the system. Whether hiring new staff or updating existing details, this feature serves as the primary control point for teacher data. Additionally, the Display All Teachers functionality offers a consolidated view of the institute's teaching workforce, aiding in resource allocation and staff management.

3. Course and Subject Management

To maintain a structured academic curriculum, the admin has full control over courses and subjects. The Add, Edit, Delete a Course and Add, Edit, Delete a Subject functionalities allow the admin to define the educational hierarchy, creating new courses and assigning relevant subjects to them. This flexibility ensures the system adapts to curriculum changes. Furthermore, the admin can verify these additions using the Display All Courses and Display All Subjects features, which provide organized lists of all academic offerings available in the institute.

4. Attendance and Academic Performance

Monitoring daily operations and academic standards is a critical responsibility. The admin utilizes the View and Update Attendance Of Teachers And Students functionality to oversee punctuality and regularity across the institute. This dual-purpose feature ensures that both staff and student attendance records are monitored from a single interface.

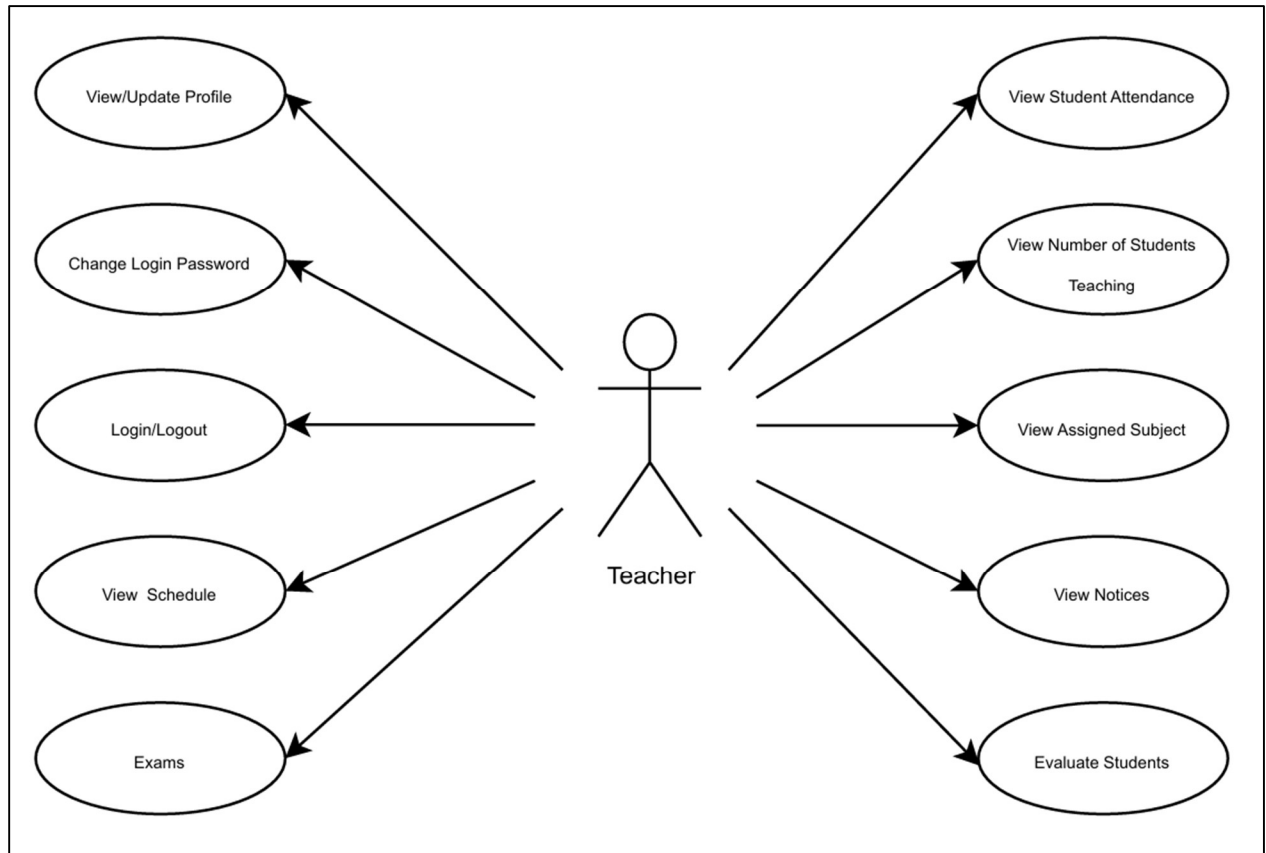
Regarding academic performance, the admin manages grading through the Add Marks of Students and View and Update Marks Of Students features. This centralized control over examination results ensures accurate report generation and allows the administration to address any discrepancies in student grading.

5. Communication, Complaints, and Profile Management

Effective communication and grievance redressal are vital for a smooth operational environment. The admin manages official announcements via Add A notice and Display All Notices functionalities, ensuring important updates reach the intended audience. Simultaneously, the View and Update Complaints of Students feature allows the admin to address student grievances systematically, fostering a supportive educational environment.

Finally, the View and Update Profile functionality allows the admin to maintain their own credentials and details, ensuring the security and accuracy of the administrator account itself.

2.2 Teacher Functionality



1. Login and Logout

The Login and Logout functionality allows teachers to securely access the Institute Management System using valid credentials. After successful authentication, teachers can perform academic and administrative tasks assigned to their role. The logout option ensures secure exit from the system, preventing unauthorized access and protecting sensitive data.

2. View and Update Profile

This module enables teachers to view and manage their personal and professional information, such as name, email address, contact details, designation, qualification, and assigned subjects. Teachers can update permitted details to keep their profile information accurate and up to date, reducing dependency on administrative staff.

3. Change Login Password

The Change Login Password feature allows teachers to update their login credentials whenever required. This functionality improves system security by encouraging periodic password changes and protecting accounts from unauthorized access.

4. View Schedule

The View Schedule module provides teachers with access to their teaching timetable, including lecture timings, subject schedules, and other academic activities. This helps teachers plan their daily work efficiently and avoid scheduling conflicts.

5. View Assigned Subjects

Using this functionality, teachers can view the subjects assigned to them by the administration. The module displays subject details along with the associated courses, enabling teachers to clearly understand their teaching responsibilities.

6. View Number of Students Teaching

This feature allows teachers to see the number of students enrolled in their assigned subjects or courses. Knowing class strength helps teachers plan lectures, assessments, classroom activities, and evaluation workloads more effectively.

7. View All Students

The View All Students module allows teachers to access a complete list of students enrolled in the institute. The list includes details such as student ID, name, course, email, date of birth, and gender. This centralized directory helps teachers quickly identify students for academic, administrative, or counseling purposes.

8. Access Student Attendance Records

Through the Student Attendance module, teachers can view and analyze attendance records of students. The system displays total classes conducted, classes attended, and attendance percentages. This helps teachers identify irregular students, take early corrective actions, and use attendance data for academic evaluation.

9. View Student Marks

The Student Marks functionality allows teachers to view marks obtained by students in various subjects and assessments. This centralized record helps teachers evaluate academic performance, identify high and low performers, and plan remedial or advanced learning activities accordingly.

10. Exams Management

The Exams module enables teachers to view and manage exam-related information such as exam schedules, subjects, total marks, and exam status. This ensures proper coordination between teaching activities and assessment processes.

11. Evaluate Students

The Evaluate Students feature allows teachers to assess student performance based on marks, attendance, and exam results. Teachers can provide feedback, identify learning gaps, and support students in improving their academic performance.

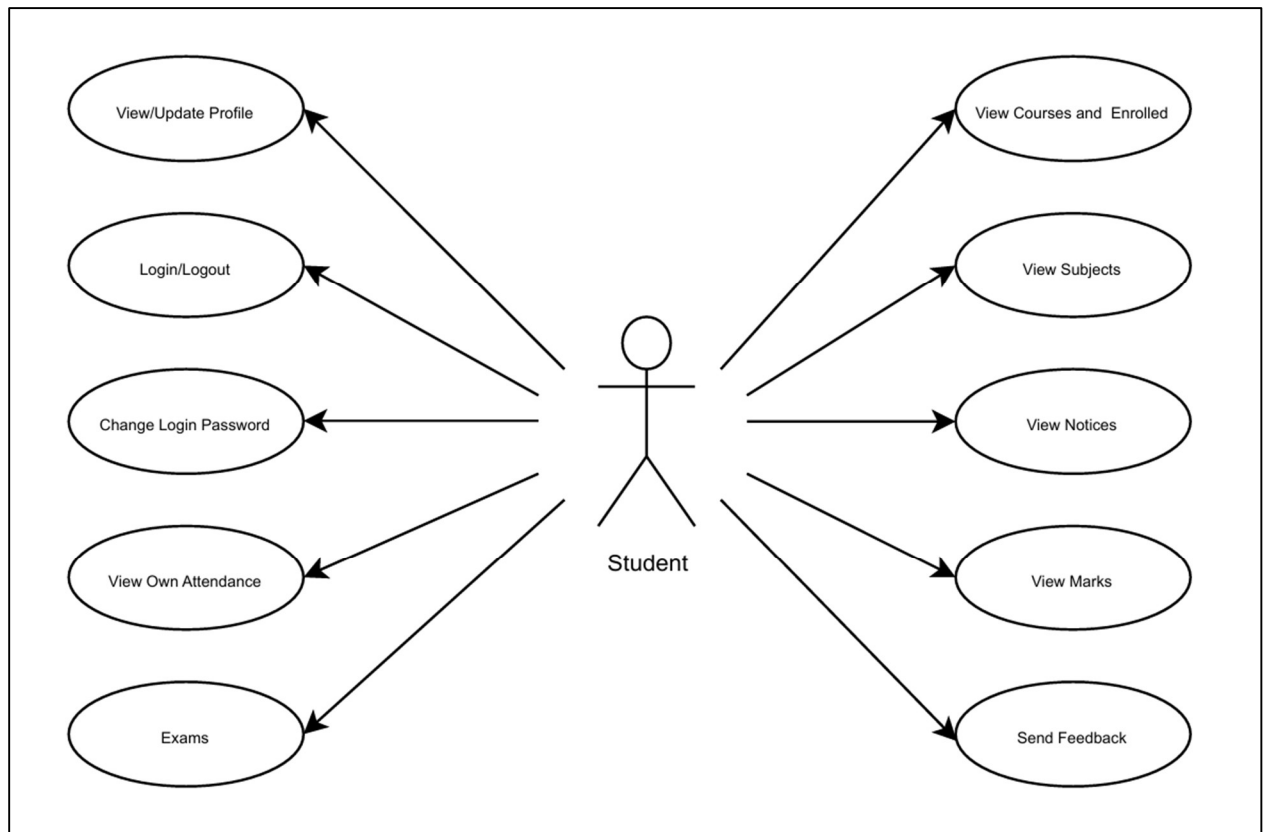
12. View Other Teachers and Courses

This module allows teachers to view details of other faculty members and courses offered by the institute. It supports collaboration among teachers, coordination of substitute classes, and better understanding of the overall academic structure.

13. View Notices

The View Notices functionality keeps teachers informed about important announcements such as exam schedules, meetings, training programs, and institutional updates. This ensures timely communication and helps teachers stay aligned with institute policies and activities.

2.3 Student Functionalities



1. Login and Logout

The Login and Logout functionality enables students to securely access the Institute Management System using their registered credentials. After successful login, students can access academic and administrative features relevant to their role. The logout option allows students to safely exit the system, ensuring account security and preventing unauthorized access to personal data.

2. Profile Management (View/Update Profile)

The Profile Management module allows students to view and update their personal and academic details stored in the system. This includes information such as full name, date of birth, contact details, email address, gender, admission date, and enrolled course. Access to profile information enables students to verify the correctness of their records and keep them updated. This reduces dependency on

administrative staff for minor corrections and ensures that all academic records are linked to accurate student information.

3. Change Login Password

The Change Login Password functionality allows students to update their account password whenever required. This feature improves system security by enabling periodic password changes and protecting student accounts from unauthorized access. It ensures the confidentiality of personal and academic information stored in the system.

4. View Courses Enrolled

Through the View Courses Enrolled feature, students can see the list of courses in which they are currently registered. The module displays course-related details such as course name, duration, and status. This functionality helps students clearly understand their academic enrollment and track the courses they are pursuing within the institute.

5. View Subjects

The View Subjects module allows students to view subjects associated with their enrolled courses. This includes subject names and related academic information. Having access to subject details helps students understand the course structure and prepare effectively for lectures, assignments, and examinations.

6. View Own Attendance

The View Own Attendance functionality enables students to check their attendance records for each subject or course. The attendance details include the total number of classes conducted, classes attended, and attendance percentage. This real-time access helps students monitor attendance regularly, identify shortages early, and take corrective action to meet eligibility requirements for exams and internal assessments.

7. View Performance Statement (View Marks)

The View Performance Statement module allows students to view their marks and overall academic performance. This includes subject-wise marks, total marks, percentages, and grades. By accessing this information, students can evaluate their academic progress, identify strengths and weaknesses, and plan improvement strategies. This feature supports self-assessment and helps students prepare for future examinations or academic opportunities.

8. Exams

The Exams module provides students with access to examination-related information such as exam schedules, exam names, dates, and total marks. This functionality ensures that students are well informed about upcoming examinations and can prepare accordingly. Centralized exam information reduces confusion and improves examination readiness.

9. View Notices

The View Notices functionality allows students to stay updated with important institute announcements, academic notifications, exam schedules, and administrative updates. Notices are displayed in a centralized digital format, ensuring timely and uniform communication. This feature helps students stay informed and aligned with institute activities and policies.

10. Feedback Management (Send Feedback)

The Feedback Management module allows students to submit feedback related to academics, faculty, courses, or institute facilities. Students can provide comments and ratings through a structured feedback form. Once submitted, the feedback is stored in a centralized database and reviewed by the administration. This feature encourages open communication, helps improve institute services, and ensures that student opinions are acknowledged without the need for physical follow-ups.

2.2 NON-FUNCTIONAL REQUIREMENTS

2.2.1 Interface

Go to Appendix B for user interfaces.

2.2.2 Performance

- **Number of Concurrent Users:**IMS shall be able to manage at least 1000 transactions/inquiries per second

2.2.3 Constraint

IMS shall be able to manage at least 1000 transactions/inquiries per second.

2.2.4 Other Requirements:

- **Hardware Interfaces**

The SPMS is expected to function on Intel PIII 900 MHz Processor equivalent or above, 128 MB RAM, 20 GB HDD.

- **Software Interfaces**

The SPMS shall work on MS Windows operating systems family (MS Windows 98, MS Windows NT Workstation, MS Windows 2000, MS Windows XP). It configures to work with Oracle database. This System works on Apache Tomcat server. It uses browser IE 5.0 & above. It uses IIS 5.0 server.

3. DESIGN

3.1 Database Design

The following table structures depict the database design.

Table1: Users

Field	Type	Null	Key	Default	Extra
user_id	bigint	NO	PRI	NULL	auto_increment
created_on	date	YES		NULL	
last_updated	datetime(6)	YES		NULL	
gender	enum('FEMALE','MALE','OTHER')	YES		NULL	
address	varchar(500)	NO		NULL	
admission_date	date	NO		NULL	
designation	varchar(100)	NO		NULL	
dob	date	NO		NULL	
email	varchar(100)	NO	UNI	NULL	
full_name	varchar(100)	NO		NULL	
last_login	datetime(6)	YES		NULL	
password	varchar(255)	NO		NULL	
phone	varchar(50)	NO	UNI	NULL	
profile_image	varchar(500)	YES		NULL	
qualification	varchar(100)	NO		NULL	
role	enum('ADMIN','ALL','STUDENT',...)	NO		NULL	
roll_number	varchar(100)	NO		NULL	
active	bit(1)	YES		NULL	
user_name	varchar(100)	NO	UNI	NULL	

Table2: Attendances

Field	Type	Null	Key	Default	Extra
attendance_id	bigint	NO	PRI	NULL	auto_increment
created_on	date	YES		NULL	
last_updated	datetime(6)	YES		NULL	
attendance_date	date	NO		NULL	
status	enum('ABSENT','LEAVE','PRESENT')	NO		NULL	
course_id	bigint	NO	MUL	NULL	
subject_id	bigint	NO	MUL	NULL	
user_id	bigint	NO	MUL	NULL	

Table3: Courses

Field	Type	Null	Key	Default	Extra
course_id	bigint	NO	PRI	NULL	auto_increment
created_on	date	YES		NULL	
last_updated	datetime(6)	YES		NULL	
description	varchar(500)	YES		NULL	
duration	varchar(255)	YES		NULL	
end_date	date	YES		NULL	
fees	double	YES		NULL	
max_students	int	YES		NULL	
name	varchar(100)	NO		NULL	
start_date	date	YES		NULL	
active	bit(1)	YES		NULL	

Table4: Subjects

Field	Type	Null	Key	Default	Extra
subject_id	bigint	NO	PRI	NULL	auto_increment
created_on	date	YES		NULL	
last_updated	datetime(6)	YES		NULL	
description	varchar(500)	YES		NULL	
duration	varchar(500)	YES		NULL	
name	varchar(100)	NO		NULL	
schedule_path	varchar(255)	YES		NULL	
active	bit(1)	YES		NULL	
course_id	bigint	NO	MUL	NULL	
teacher_id	bigint	YES	MUL	NULL	

Table5: Marks

Field	Type	Null	Key	Default	Extra
mark_id	bigint	NO	PRI	NULL	auto_increment
created_on	date	YES		NULL	
last_updated	datetime(6)	YES		NULL	
obtained_marks	int	YES		NULL	
status	bit(1)	YES		NULL	
exam_id	bigint	YES	MUL	NULL	
student_id	bigint	YES	MUL	NULL	

Table6: Notices

Field	Type	Null	Key	Default	Extra
notice_id	bigint	NO	PRI	NULL	auto_increment
created_on	date	YES		NULL	
last_updated	datetime(6)	YES		NULL	
description	text	YES		NULL	
publish_date	date	YES		NULL	
active	bit(1)	YES		NULL	
target_role	enum('ADMIN','ALL','STUDENT','TEACHER')	YES		NULL	
title	varchar(200)	YES		NULL	

Table7: Enrollments

Field	Type	Null	Key	Default	Extra
enrollments_id	bigint	NO	PRI	NULL	auto_increment
created_on	date	YES		NULL	
last_updated	datetime(6)	YES		NULL	
assigned_date	date	YES		NULL	
is_inactive	bit(1)	YES		NULL	
role	enum('ADMIN','ALL','STUDENT','TEACHER')	NO		NULL	
course_id	bigint	YES	MUL	NULL	
subject_id	bigint	YES	MUL	NULL	
user_id	bigint	YES	MUL	NULL	

Table8: Exams

Field	Type	Null	Key	Default	Extra
exam_id	bigint	NO	PRI	NULL	auto_increment
created_on	date	YES		NULL	
last_updated	datetime(6)	YES		NULL	
exam_date	date	YES		NULL	
exam_link	varchar(500)	YES		NULL	
exam_name	varchar(100)	YES		NULL	
status	bit(1)	YES		NULL	
total_marks	int	YES		NULL	
course_id	bigint	YES	MUL	NULL	
subject_id	bigint	YES	MUL	NULL	

Table9: Feedback

Field	Type	Null	Key	Default	Extra
feedback_id	bigint	NO	PRI	NULL	auto_increment
created_on	date	YES		NULL	
last_updated	datetime(6)	YES		NULL	
feedback_date	date	YES		NULL	
feedback_text	varchar(500)	NO		NULL	
is_inactive	bit(1)	YES		NULL	
rating	int	NO		NULL	
responded_at	datetime(6)	YES		NULL	
response_text	varchar(500)	YES		NULL	
course_id	bigint	YES	MUL	NULL	
subject_id	bigint	YES	MUL	NULL	
user_id	bigint	YES	MUL	NULL	

E-R Diagram,Dataflow diagram and Class Diagram:

Go to Appendix A

4. CODING STANDARDS IMPLEMENTED

Naming and Capitalization

Below summarizes the naming recommendations for identifiers in Pascal casing is used mainly (i.e. capitalize first letter of each word) with camel casing (capitalize each word except for the first one) being used in certain circumstances.

Identifier	Case	Examples	Additional Notes
Class	Pascal	Person, BankVault, SMSMessage, Dept	Class names should be based on "objects" or "real things" and should generally be nouns . No ‘_’ signs allowed. Do not use type prefixes like ‘C’ for class.
Method	Camel	getDetails, updateStore	Methods should use verbs or verb phrases.
Parameter	Camel	personName, bankCode	Use descriptive parameter names. Parameter names should be descriptive enough that the name of the parameter and its type can be used to determine its meaning in most scenarios.
Interface	Pascal with "I" prefix	Disposable	Do not use the ‘_’ sign
Property	Pascal	ForeColor, BackColor	Use a noun or noun phrase to name properties.
Associated private member variable	_camelCase	_foreColor, _backColor	Use underscore camel casing for the private member variables
Exception Class	Pascal with "Exception" suffix	WebException,	

Comments

- Comment each type, each non-public type member, and each region declaration.
- Use end-line comments only on variable declaration lines. End-line comments are comments that follow code on a single line.
- Separate comments from comment delimiters (apostrophe) or // with one space.
- Begin the comment text with an uppercase letter.
- End the comment with a period.
- Explain the code; do not repeat it.

5. TEST REPORT

Another group called Linux did the testing and the report of the testing is given hereunder.

GENERAL TESTING:

SR - N O	TEST CASE	EXPECTED RESULT	ACTUAL RESULT	ERROR MESSAGE
1	Register Page	Redirected to Next page	OK	Nothing
2	Login Page	Pop-up will come	Ok	Please enter username and password again.
3	Dashboard	Count of courses, students, teachers, latest notices, details of topper	Ok	Nothing
4	Student functionaliti es	Redirect to next: Add, display, attendance, marks, fees of students	Ok	Nothing
5	Teacher Functionalit ies	Redirect to next: Add, display, attendance of teachers	Ok	Nothing
6	Course	Redirect to next: Add, display all courses	Ok	Nothing
7	Notices	Redirect to next: Add, display all notices	Ok	Nothing

8	Subjects	Redirect to next: Add, display all subjects	Ok	Nothing
9	Complaints	Redirect to next: Display and update all complaints	Ok	Nothing
10	Profile	View, update profile	Ok	Nothing
11	Marks	Redirect to next: Add, update student marks	Ok	Nothing
12	Logout	It will log out from user profile.	Ok	Nothing

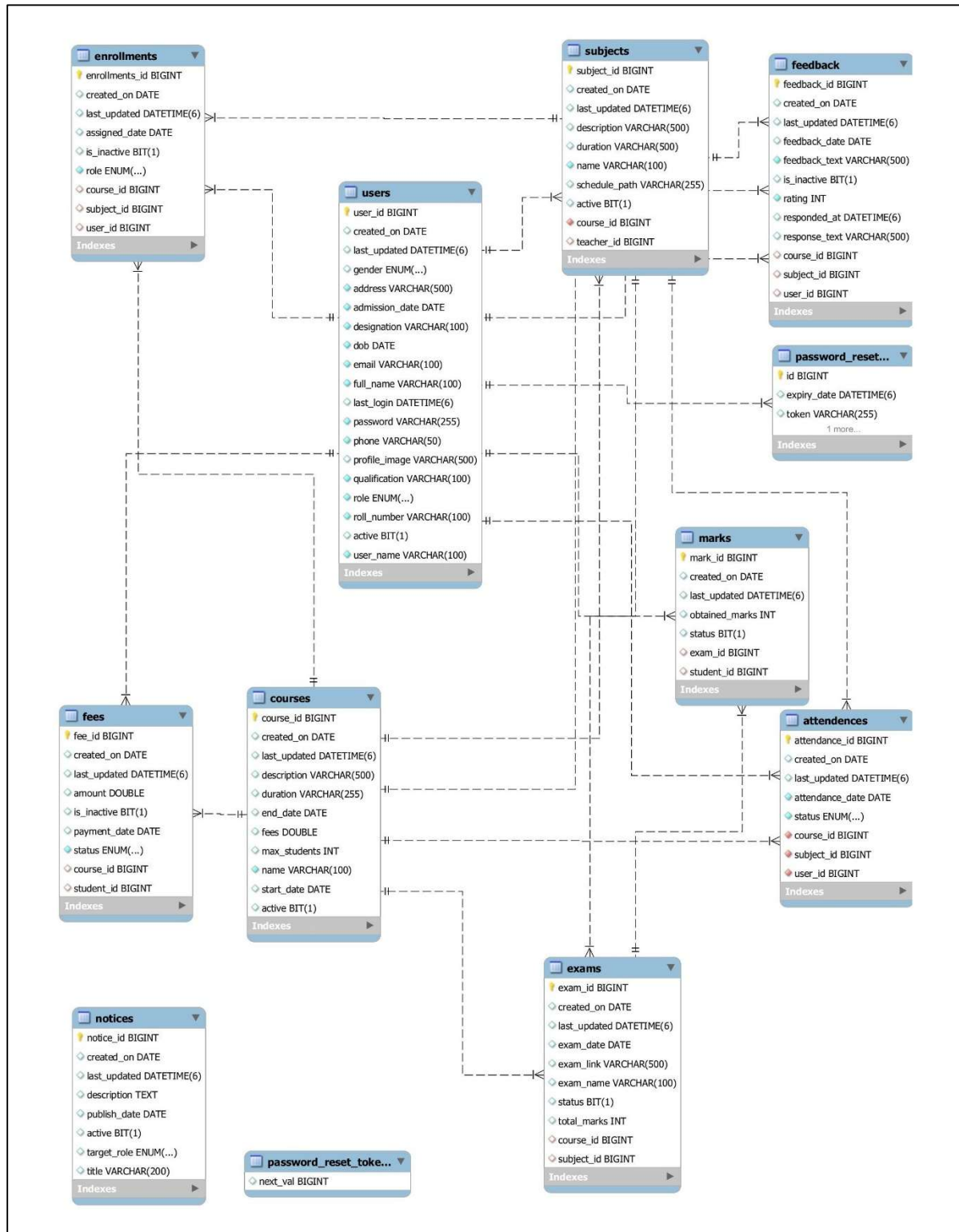
6. PROJECT MANAGEMENT RELATED STATISTICS

DATE	WORK PERFORMED		Additional Notes
DEC 21, 2025	Project Allotment and User Requirements Gathering	Feasibility Study	Our team met the client Mr. Nitin kudale (CEO, SIIT Pune) to know his requirements.
DEC 22, 2025	Initial SRS Document Validation And Team Structure Decided	Requirement Analysis (Elicitation)	The first SRS was presented to the client to understand his requirements better
DEC 23, 2025	Designing the use-cases, Class Diagram, Collaboration Diagram, E-R Diagram and User Interfaces	Requirement Analysis & Design Phase	Database Design completed.
DEC 24, 2025	Business Logic Component Design Started	Design Phase	-----
DEC 26-28,2025	Coding Phase Started	Coding Phase	70% of Class Library implemented.
DEC 29,2025	Implementation of Web Application and Window Application Started	Coding Phase	Class Library Development going on.
JAN 3,2026	Off	Off	Off
JAN 5-6,2026	Implementation of Web Application and Window Application Continued	Coding Phase and Unit Testing	Class Library Modified as per the need.

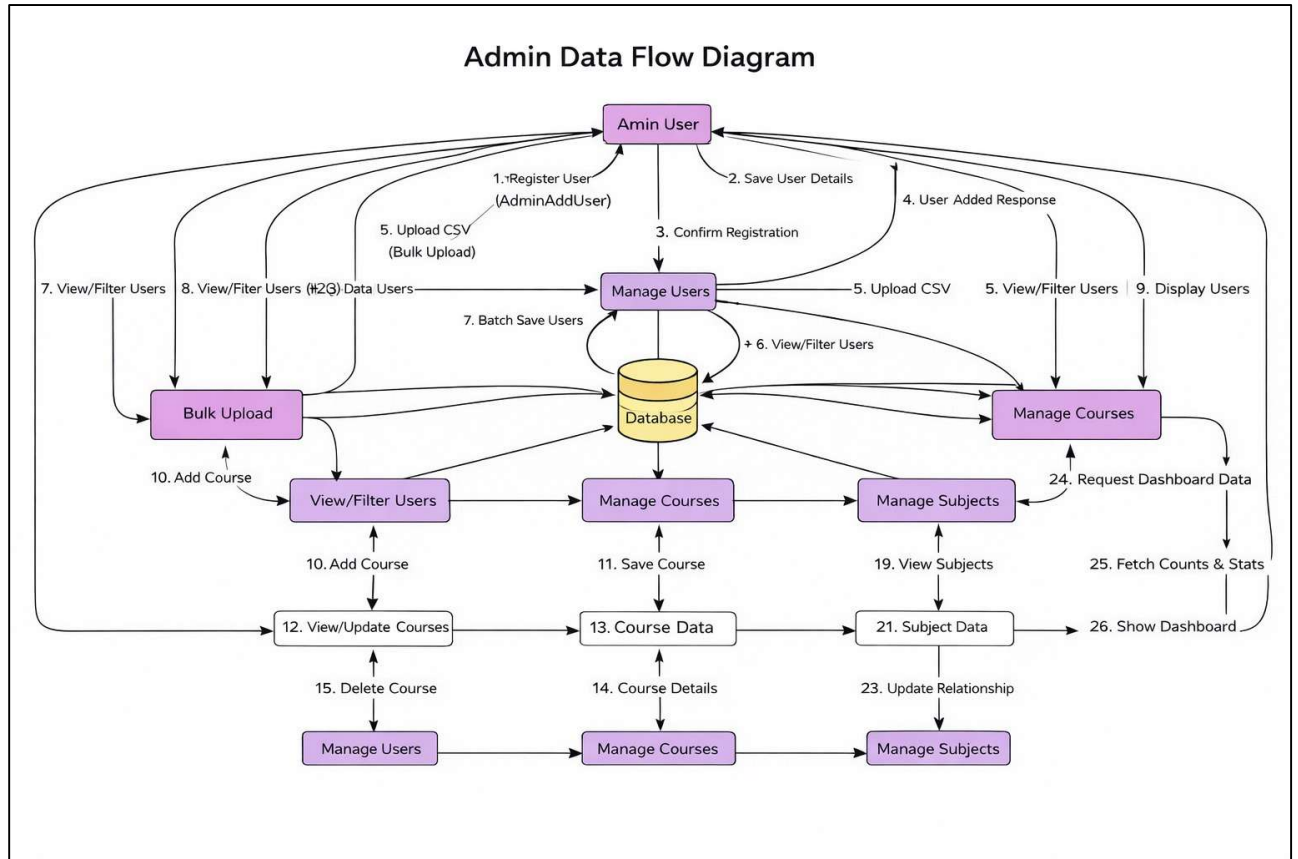
JAN 8-9,2026	Implementation of Web Application and Window Application Continued	Coding Phase and Unit Testing	--
JAN 19-20,2026	After Ensuring Proper Functioning the Required Validations were Implemented	Coding Phase and Unit Testing	Module Integration was done by the Project Manager
JAN 22 ,2026	The Project was Tested by the respective Team Leaders and the Project Manager	Testing Phase (Module Testing)	--
JAN 24,2026	The Project was Submitted to Other Project Leader of Other Project Group for Testing	Testing Phase (Acceptance Testing)	The Project of Other Team was Taken up by the Team for Testing
JAN 28-31,2026	The Errors Found were Removed.	Debugging	The Project was complete for submission
FEB 3,2026	Final Submission of Project		

Appendix A

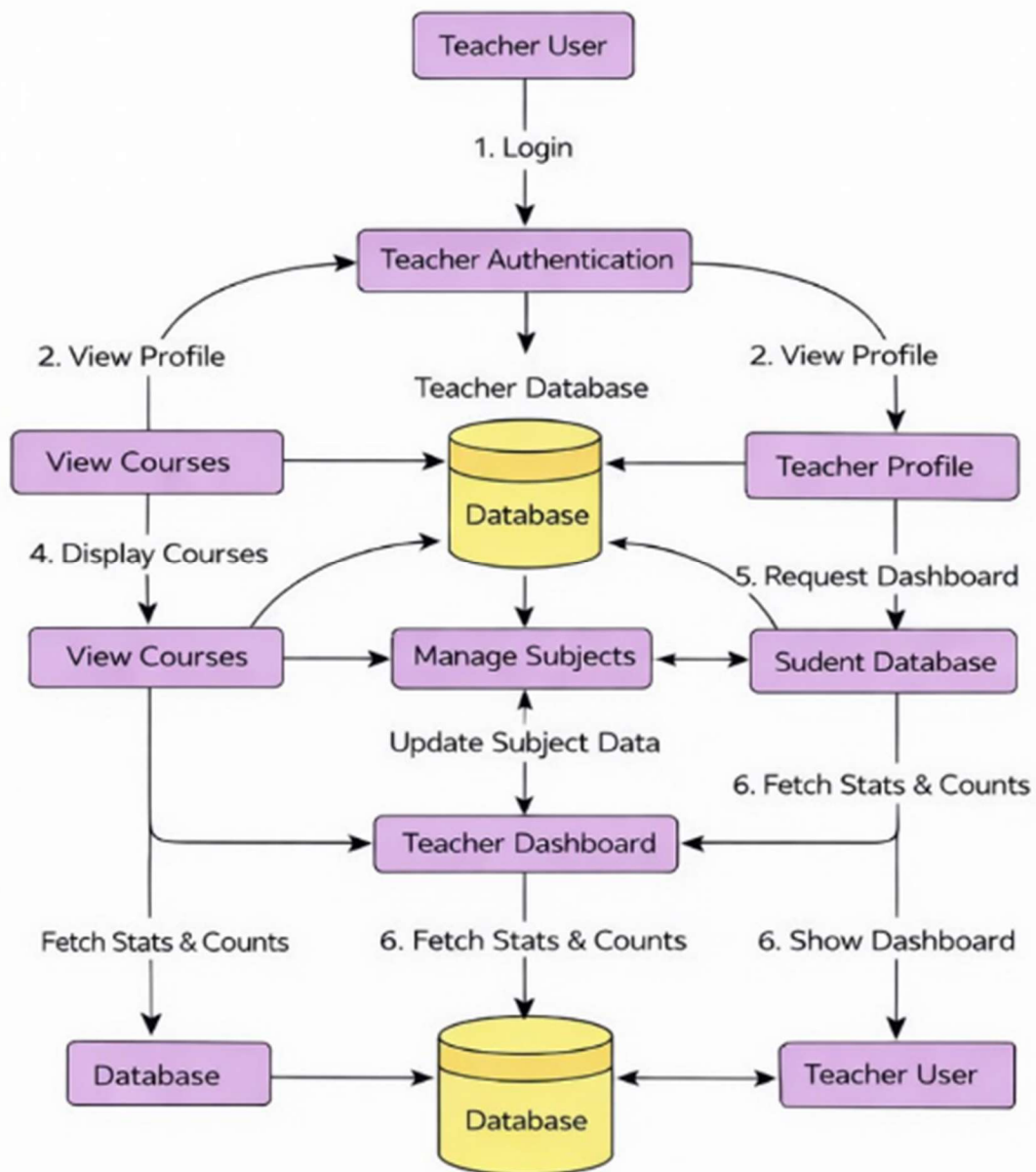
Entity Relationship Diagram



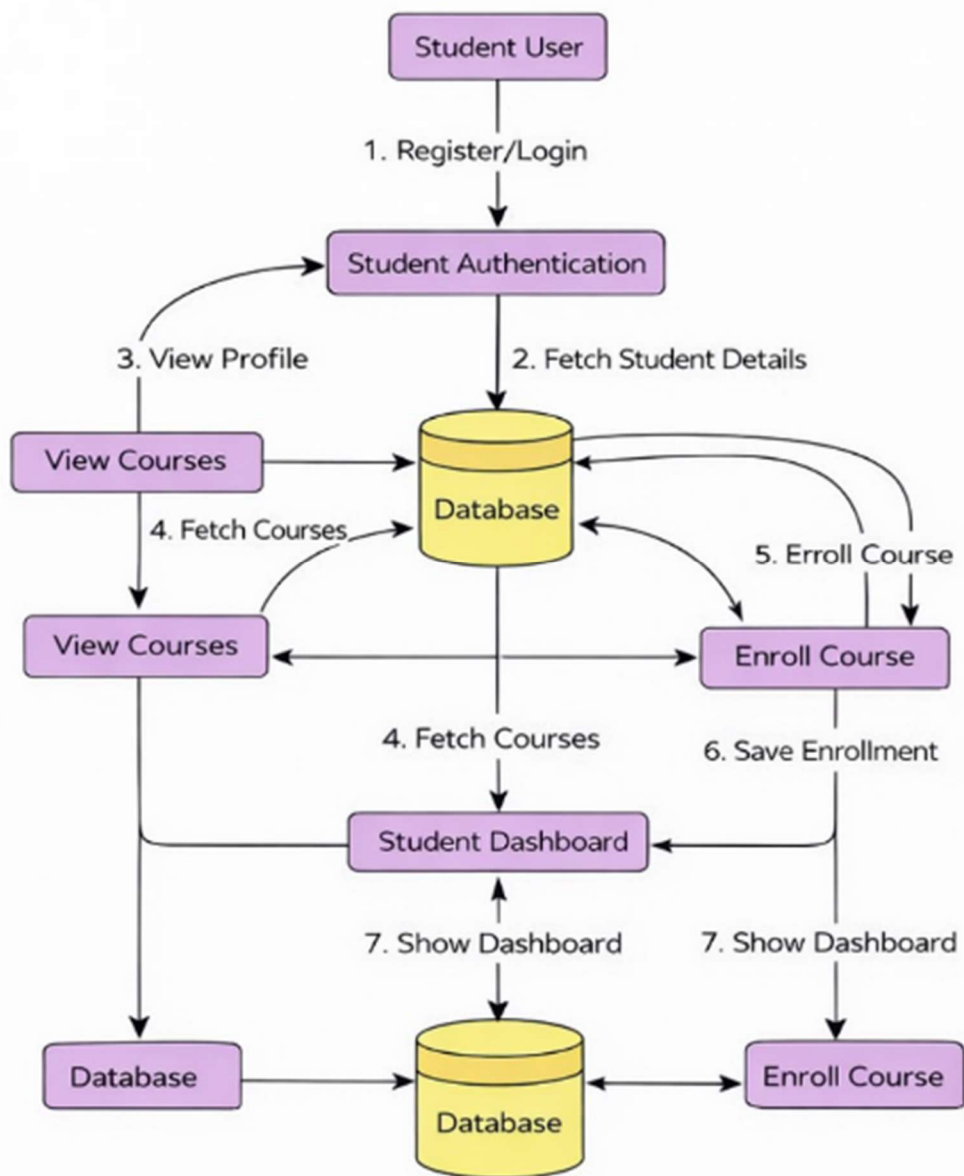
Data Flow Diagram:



Teacher Data Flow Diagram

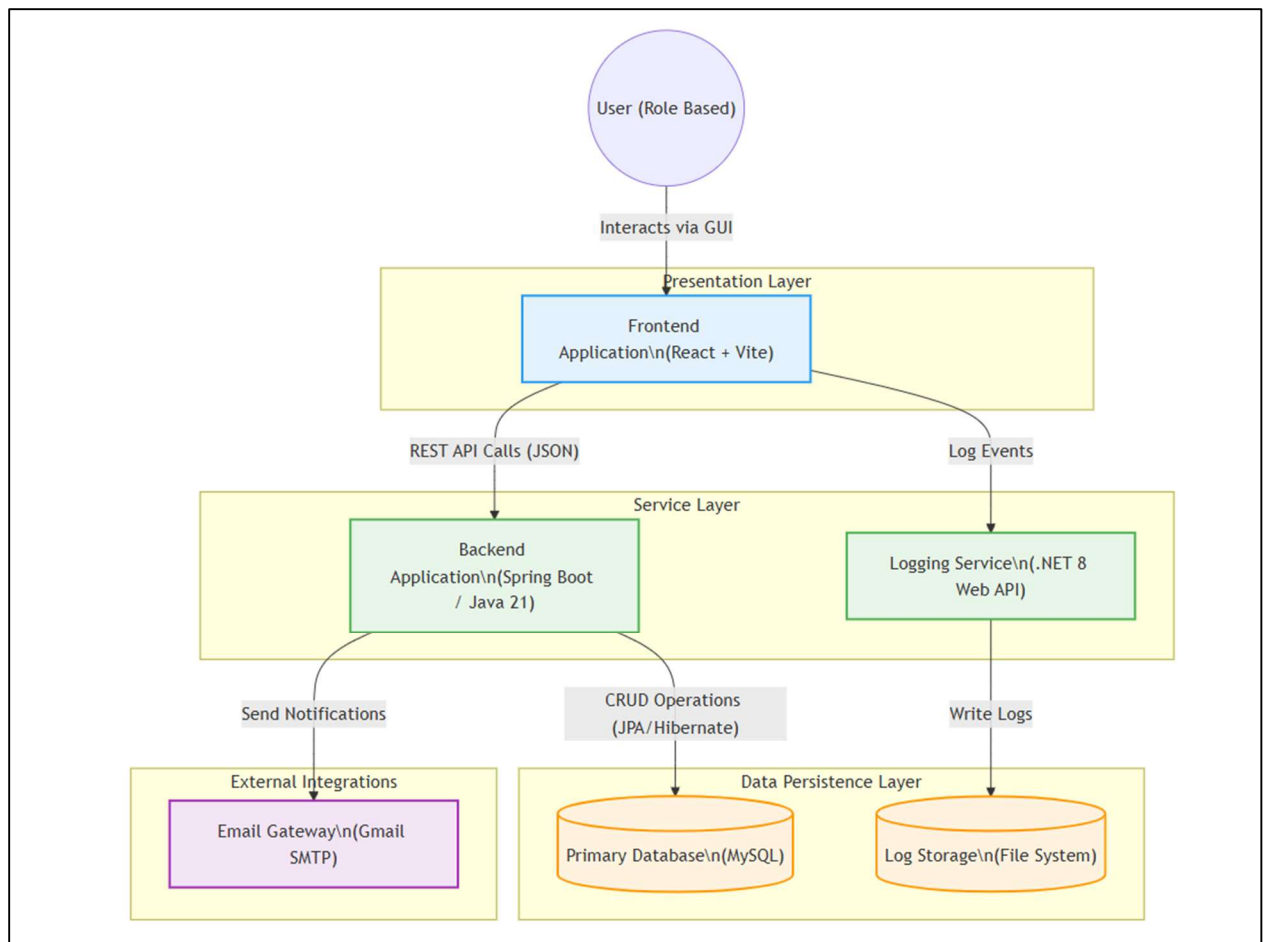


Student Data Flow Diagram



System Architecture:

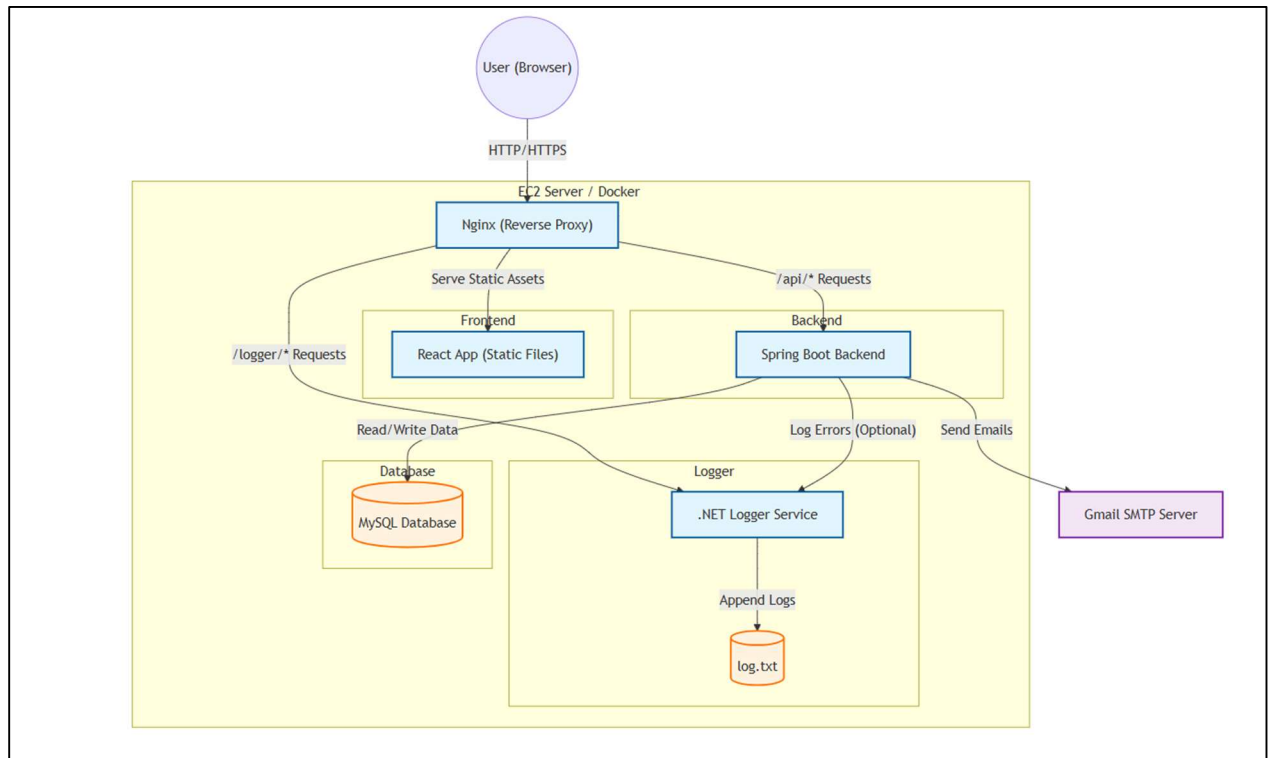
System architecture (Without Deployment):



Architectural Layers :

- **Presentation Layer:**
The client-side React application responsible for the View and User Interaction. It is role-aware (Student, Teacher, Admin).
- **Service Layer:**
Contains the business logic.
 - Spring Boot: Main API processing, authentication, and core logic.
 - .NET Service: Decoupled microservice for handling logging operations.
- **Data Persistence Layer:**
 - MySQL: Relational storage for structured business data.
 - Log Storage: File-based storage for application logs.

System architecture (With Deployment):



The Institute Management System is designed as a modular, containerized application deployed on a single AWS EC2 instance using Docker Compose. The architecture ensures separation of concerns, scalability, and ease of deployment.

1. Nginx (Reverse Proxy)

Acts as the single entry point. It serves the static React frontend files and routes API requests to the appropriate backend services (Spring Boot or .NET Logger).

2. Backend (Spring Boot)

The core application logic running on Java 21. It handles authentication (JWT), complex business rules, and communicates with the database and external email services.

3. Frontend (React)

A responsive SPA (Single Page Application) built with React and Vite. It consumes the APIs provided by the backend.

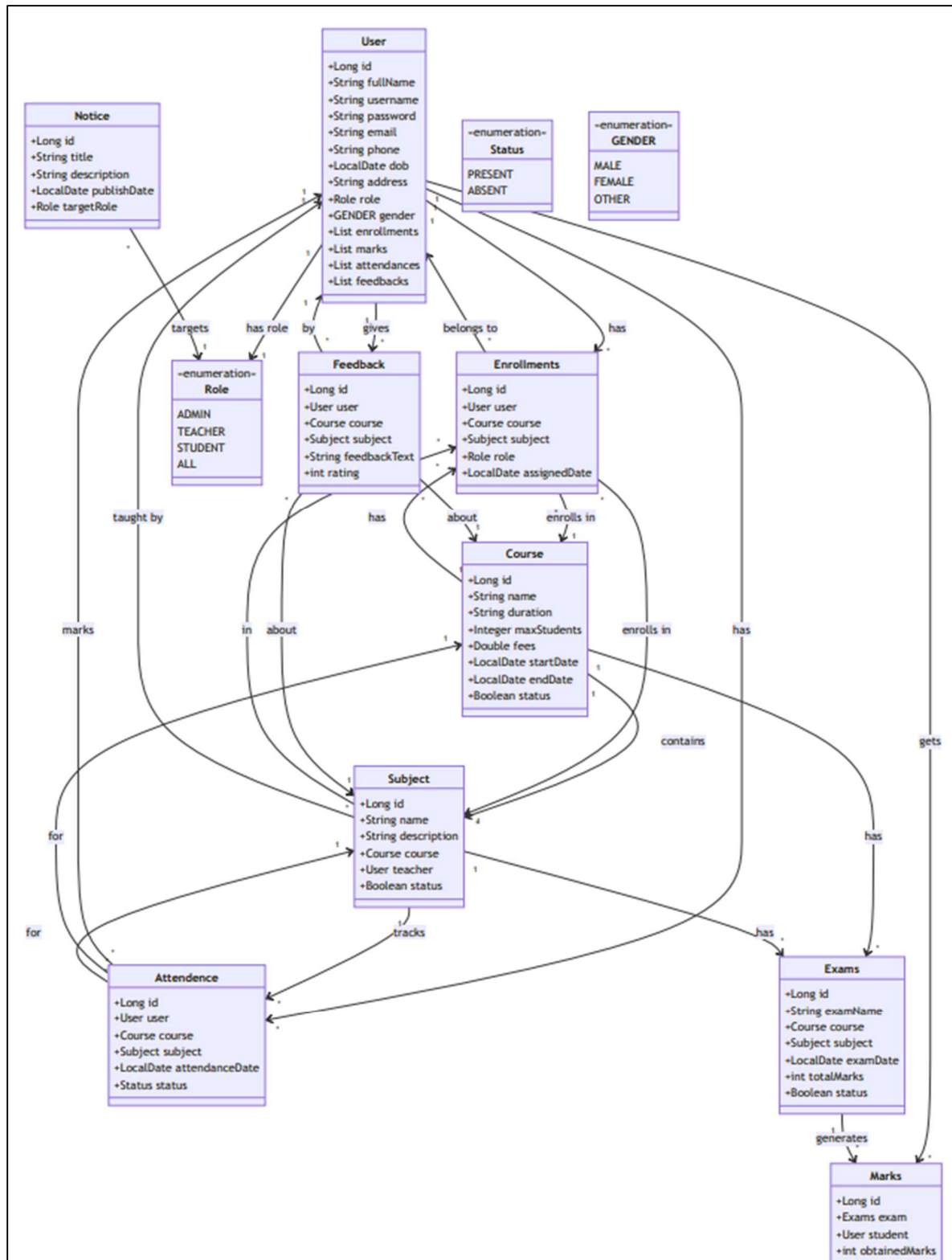
4. Database (MySQL)

A containerized relational database storing all persistent data including users, courses, and attendance records.

5. Logger Service (.NET)

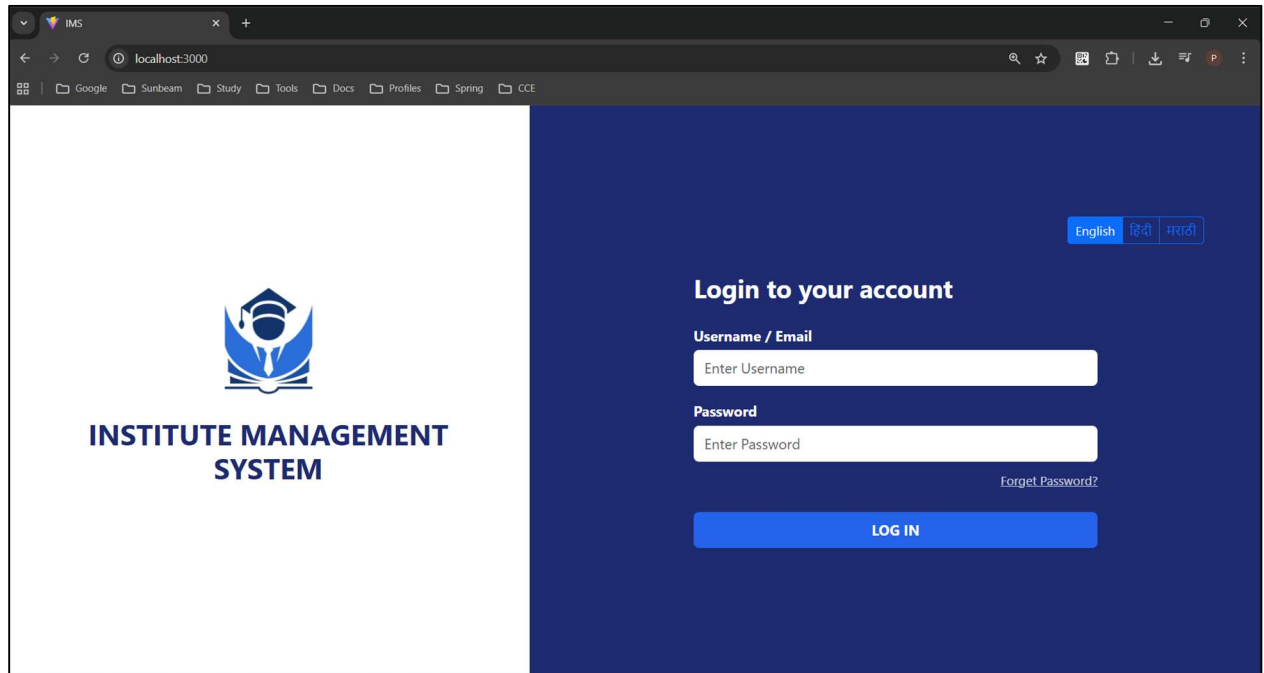
A specialized microservice built with .NET 8 for handling high-volume logging operations independently of the main backend.

Class Diagram:



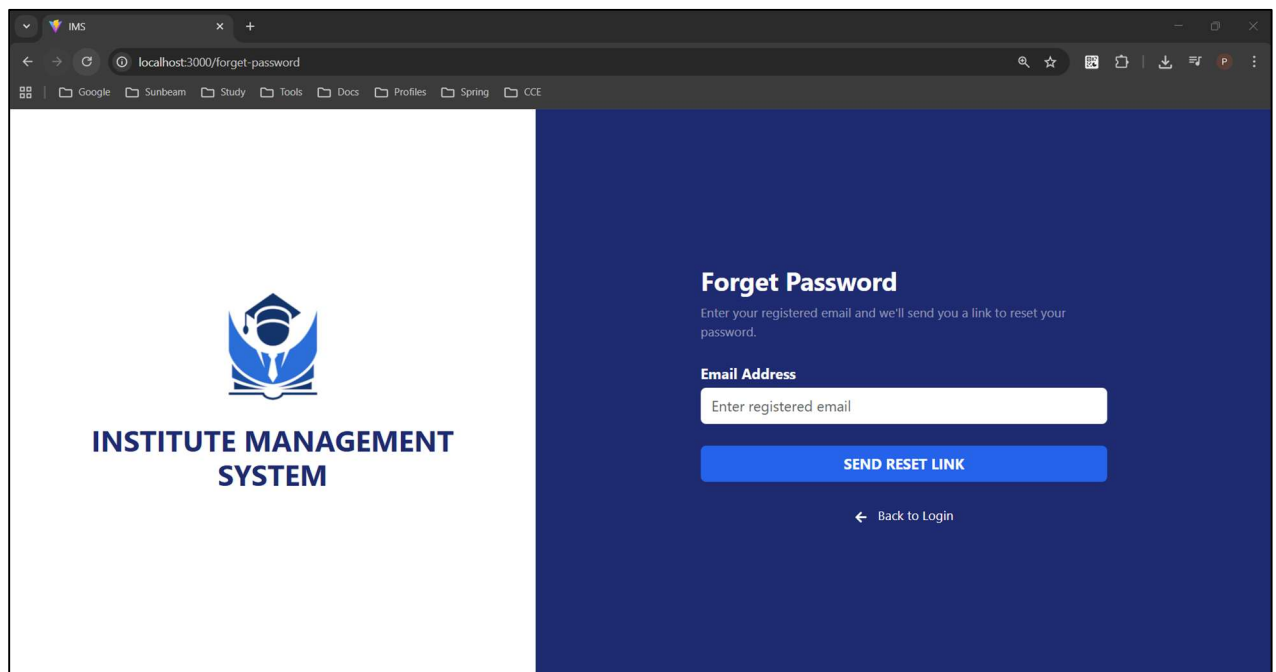
Appendix B

Login Page:



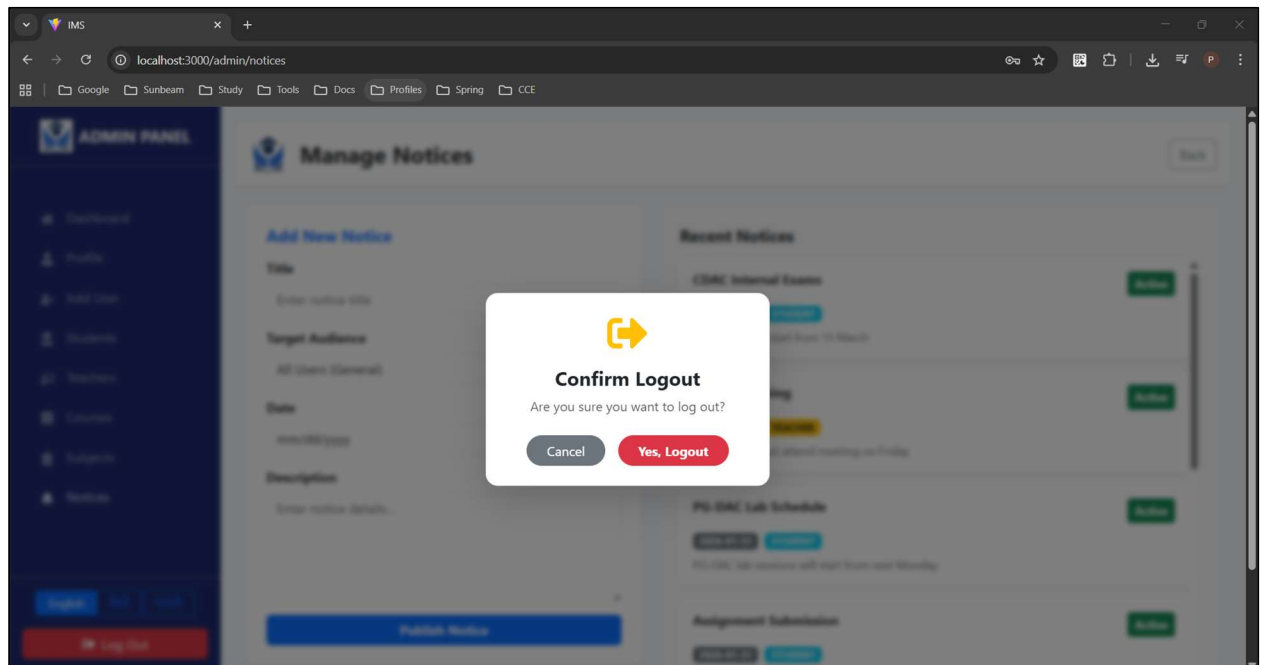
The screenshot shows a web browser window with the URL `localhost:3000`. The page is split into two main sections. On the left, there is a white background with a blue logo of a graduation cap and an open book, with the text "INSTITUTE MANAGEMENT SYSTEM" below it. On the right, there is a dark blue background. At the top right of the blue section are three language buttons: "English" (highlighted), "हिंदी", and "मराठी". Below these is the heading "Login to your account". Underneath is a form with two input fields: "Username / Email" with the placeholder "Enter Username", and "Password" with the placeholder "Enter Password". To the right of the password field is a link "Forget Password?". At the bottom of the form is a blue button labeled "LOG IN".

Forget Password:



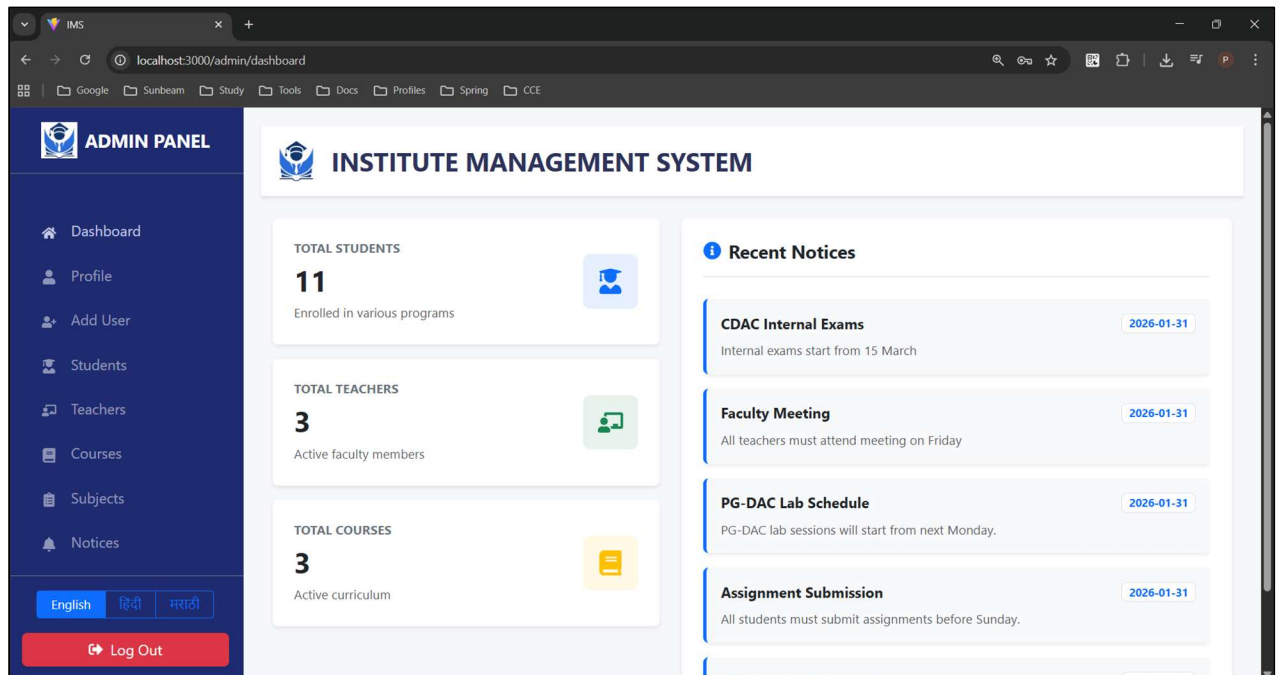
The screenshot shows a web browser window with the URL `localhost:3000/forget-password`. The layout is similar to the login page. The left side features the same white background with the blue logo and the text "INSTITUTE MANAGEMENT SYSTEM". The right side has a dark blue background. At the top right, there are no language buttons. The heading is "Forget Password". Below the heading is a paragraph: "Enter your registered email and we'll send you a link to reset your password." Underneath is a form with one input field: "Email Address" with the placeholder "Enter registered email". Below the input field is a blue button labeled "SEND RESET LINK". At the bottom of the form is a link with a left arrow and the text "Back to Login".

Logout:

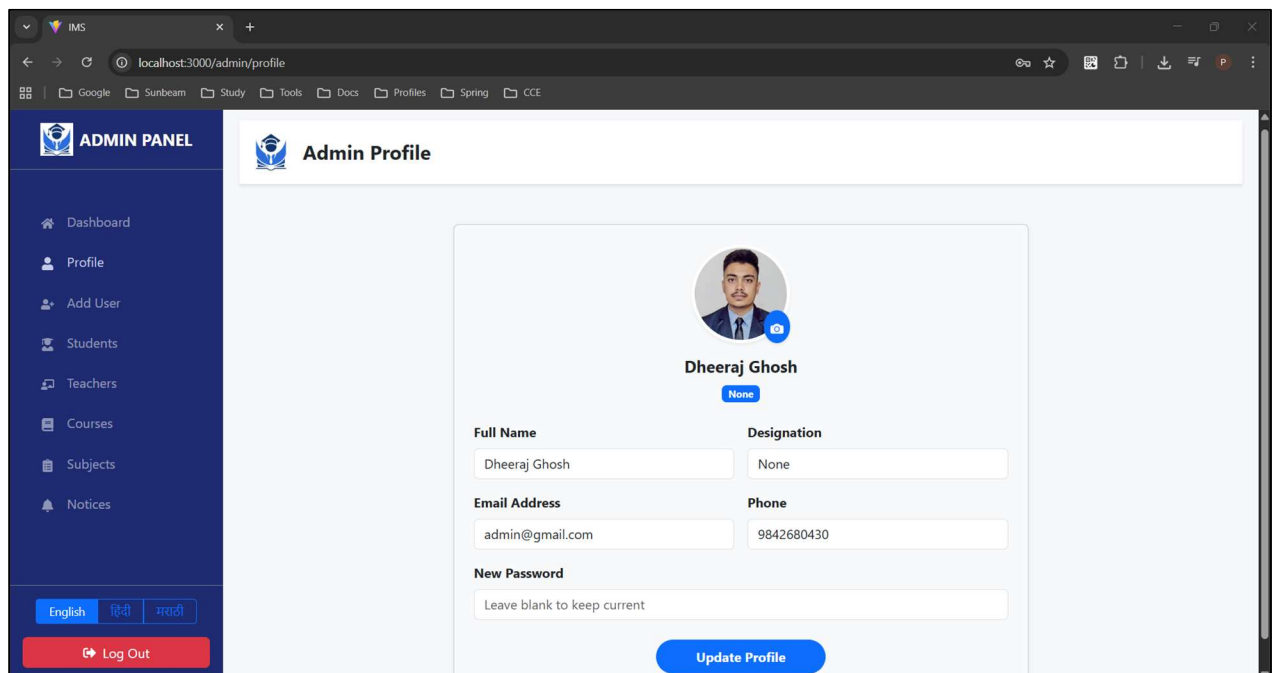


Admin

Dashboard:



Profile:



Add User: Add Single User:

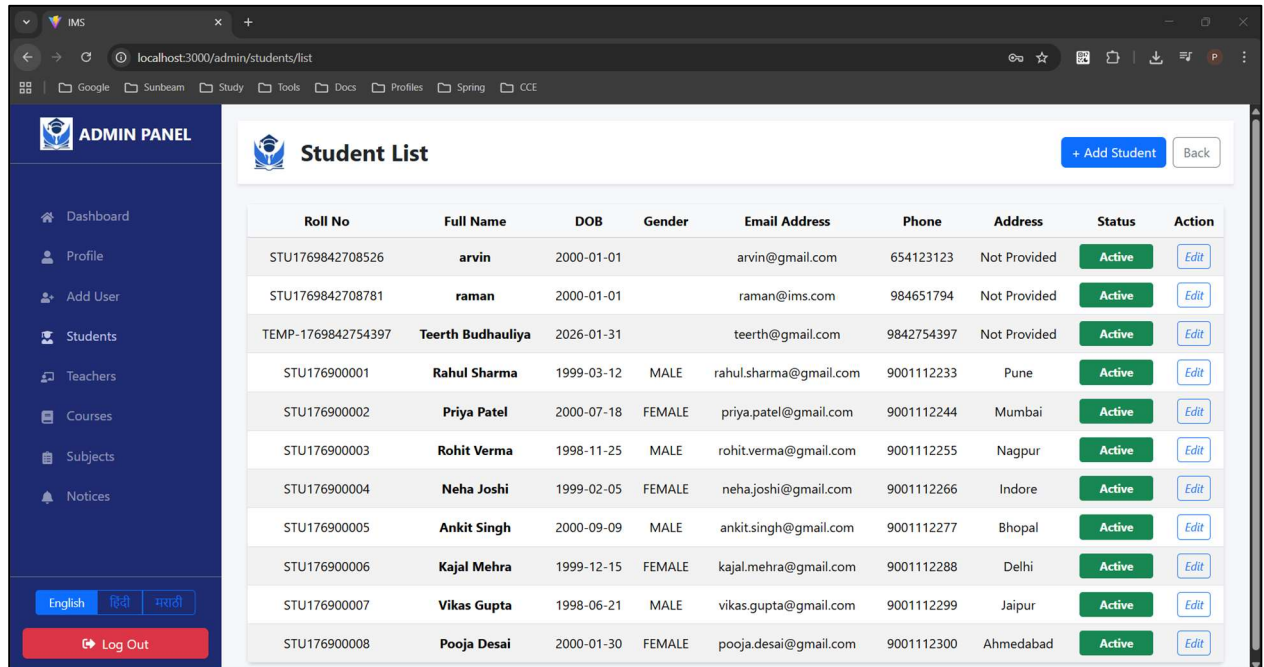
The screenshot shows a web browser at `localhost:3000/admin/add-user`. The left sidebar is the 'ADMIN PANEL' with links to Dashboard, Profile, Add User, Students, Teachers, Courses, Subjects, and Notices. At the bottom of the sidebar are language buttons (English, हिंदी, मराठी) and a Log Out button. The main content area is titled 'Add New User' and has a 'Back' button. It features two tabs: 'Single User' (selected) and 'Bulk Upload (CSV)'. The 'Single User' tab contains a form titled 'USER INFORMATION' with the following fields: 'Full Name' (text input), 'Email Address' (text input), 'Assign Role' (dropdown menu with 'Student' selected), and 'Initial Password' (text input with placeholder 'Set Password'). A blue 'Create Account' button is at the bottom of the form.

Add Multiple Users:

The screenshot shows the same web browser at `localhost:3000/admin/add-user`. The 'ADMIN PANEL' sidebar is identical. The main content area is titled 'Add New User' with a 'Back' button. The 'Bulk Upload (CSV)' tab is selected. It contains a form titled 'UPLOAD CSV FILE' with a 'Select CSV File' section. This section has a 'Choose File' button and a text area showing 'No file chosen'. Below this, the format is specified as 'Format: fullName,email,password,phone,role'. A blue 'Upload Users' button is at the bottom of the form. A light blue note box contains the following information: 'Note: Header row is required: fullName,email,password,phone,role', 'Role must be one of: STUDENT, TEACHER, ADMIN', and 'Duplicates emails will be skipped.'

Students:

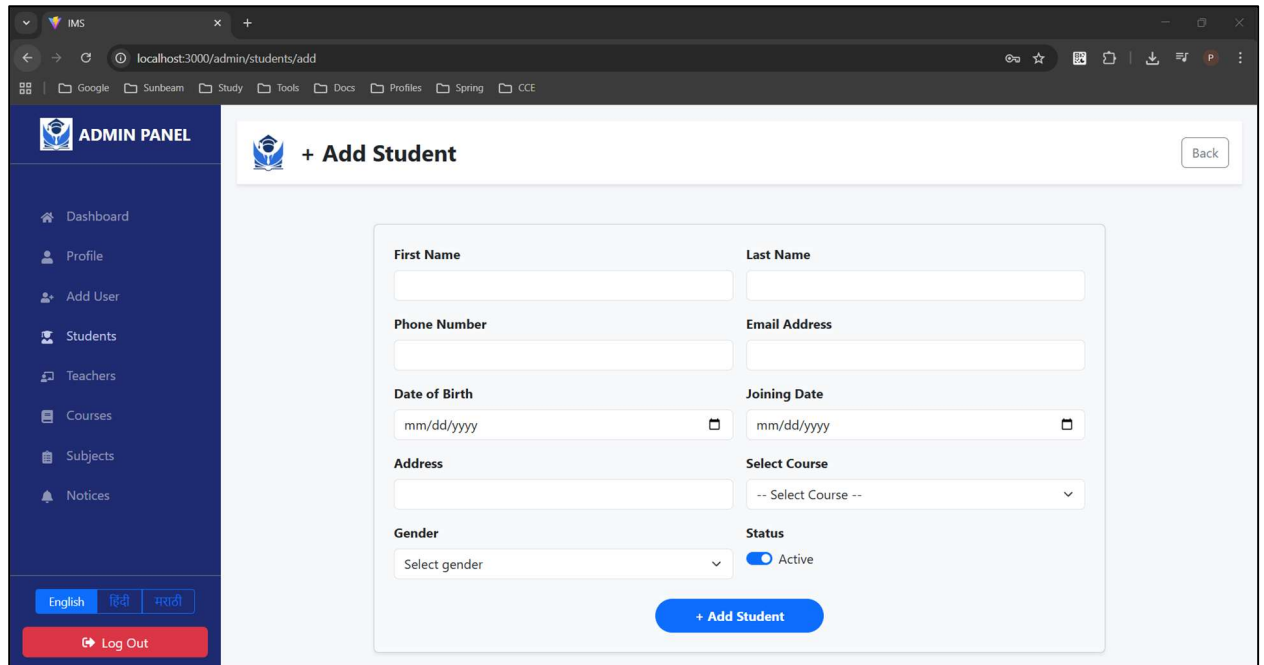
View Student:



The screenshot shows the IMS Admin Panel. The sidebar on the left contains the following links: Dashboard, Profile, Add User, Students, Teachers, Courses, Subjects, and Notices. At the bottom of the sidebar are language selection buttons for English, हिंदी, and मराठी, and a Log Out button. The main content area is titled 'Student List' and includes a '+ Add Student' button and a 'Back' button. Below these is a table with the following columns: Roll No, Full Name, DOB, Gender, Email Address, Phone, Address, Status, and Action. The table contains 11 rows of student data.

Roll No	Full Name	DOB	Gender	Email Address	Phone	Address	Status	Action
STU1769842708526	arvin	2000-01-01		arvin@gmail.com	654123123	Not Provided	Active	Edit
STU1769842708781	raman	2000-01-01		raman@ims.com	984651794	Not Provided	Active	Edit
TEMP-1769842754397	Teerth Budhauya	2026-01-31		teerth@gmail.com	9842754397	Not Provided	Active	Edit
STU176900001	Rahul Sharma	1999-03-12	MALE	rahul.sharma@gmail.com	9001112233	Pune	Active	Edit
STU176900002	Priya Patel	2000-07-18	FEMALE	priya.patel@gmail.com	9001112244	Mumbai	Active	Edit
STU176900003	Rohit Verma	1998-11-25	MALE	rohit.verma@gmail.com	9001112255	Nagpur	Active	Edit
STU176900004	Neha Joshi	1999-02-05	FEMALE	neha.joshi@gmail.com	9001112266	Indore	Active	Edit
STU176900005	Ankit Singh	2000-09-09	MALE	ankit.singh@gmail.com	9001112277	Bhopal	Active	Edit
STU176900006	Kajal Mehra	1999-12-15	FEMALE	kajal.mehra@gmail.com	9001112288	Delhi	Active	Edit
STU176900007	Vikas Gupta	1998-06-21	MALE	vikas.gupta@gmail.com	9001112299	Jaipur	Active	Edit
STU176900008	Pooja Desai	2000-01-30	FEMALE	pooja.desai@gmail.com	9001112300	Ahmedabad	Active	Edit

Add Student:



The screenshot shows the IMS Admin Panel with the '+ Add Student' form. The sidebar is the same as in the previous screenshot. The main content area is titled '+ Add Student' and includes a 'Back' button. The form contains the following fields:

- First Name (text input)
- Last Name (text input)
- Phone Number (text input)
- Email Address (text input)
- Date of Birth (date picker, format mm/dd/yyyy)
- Joining Date (date picker, format mm/dd/yyyy)
- Address (text input)
- Select Course (dropdown menu, options: -- Select Course --)
- Gender (dropdown menu, options: Select gender)
- Status (radio button, options: Active)

At the bottom of the form is a '+ Add Student' button.

Add Marks:

ADMIN PANEL

Dashboard
Profile
Add User
Students
Teachers
Courses
Subjects
Notices

English ગુજરાતી मराठी

Log Out

Student Marks

Back

Select Course -- Select Subject -- Search

Roll No	Student Info	Obtained Marks	total_marks	exam_date	Status
No marks found or select criteria to search.					

Feedback:

ADMIN PANEL

Dashboard
Profile
Add User
Students
Teachers
Courses
Subjects
Notices

English ગુજરાતી मराठी

Log Out

Student Feedbacks

Back

Student Details	Courses / Subjects	Feedback	Response	Rating	Date	Status	Action
arvin ID: 2	PG-DAC (Core Java)	Excellent teaching and content	—	★★★★★	2026-01-31	Pending	Respond
raman ID: 5	PG-DAC (Spring Boot)	Need more hands-on labs	—	★★★★	2026-01-31	Pending	Respond

Student Attendance:

The screenshot shows the 'Student Attendance' page in the IMS Admin Panel. The left sidebar contains the 'ADMIN PANEL' with links to Dashboard, Profile, Add User, Students, Teachers, Courses, Subjects, and Notices. At the bottom of the sidebar are language buttons (English, हिंदी, मराठी) and a Log Out button. The main content area is titled 'Student Attendance' and includes a 'Back' button and a dropdown menu set to 'All Courses'. Below this is a table with the following data:

Roll No	Student Info	Courses	Attendance (%)	Status
STU1769842708526	arvin	PG-DAC	100%	Good
STU1769842708526	arvin	PG-DBDA	%	Low
STU1769842708781	raman	PG-DAC	0%	Low
STU1769842708781	raman	PG-DESD	%	Low
TEMP-1769842754397	Teerth Budhauriya	PG-DBDA	100%	Good
TEMP-1769842754397	Teerth Budhauriya	PG-DAC	%	Low

Teachers:

View Teacher:

The screenshot shows the 'Teachers List' page in the IMS Admin Panel. The left sidebar is identical to the previous page. The main content area is titled 'Teachers List' and includes a '+ Add Teacher' button and a 'Back' button. Below this is a table with the following data:

ID	Full Name	Joining Date	Gender	Email Address	Mobile	Qualification	Address	Status	Actions
3	Siddhesh	2026-01-31		sid@ims.com	69333232	N/A	Not Provided	Active	Edit
4	manish	2026-01-31		manish@ims.com	32165412	N/A	Not Provided	Active	Edit
6	Prathamesh Pimpalkar	2026-01-31	MALE	prathameshpimpalkar2@gmail.com	98427328867	Not Provided	Not Provided	Active	Edit

Add Teacher:

ADMIN PANEL

- Dashboard
- Profile
- Add User
- Students
- Teachers
- Courses
- Subjects
- Notices

English हिंदी मराठी

Log Out

+ Add Teacher

Back

First Name	Last Name
Mobile	Email Address
Date of Birth	Joining Date
Address	Qualification
Assign Course	Assign Subject
Gender	Status

+ Add Teacher

View Attendance:

ADMIN PANEL

- Dashboard
- Profile
- Add User
- Students
- Teachers
- Courses
- Subjects
- Notices

English हिंदी मराठी

Log Out

Teacher Attendance

Back

ID	Teachers	Attendance (%)	Status
4	manish	%	Needs Improve
6	Prathamesh Pimpalkar	%	Needs Improve
3	Siddhesh	%	Needs Improve

Courses:

The screenshot shows the 'Course List' page in the IMS Admin Panel. The left sidebar contains the 'ADMIN PANEL' menu with options: Dashboard, Profile, Add User, Students, Teachers, Courses, Subjects, and Notices. At the bottom of the sidebar are language buttons for English, हिंदी, and मराठी, and a Log Out button. The main content area is titled 'Course List' and includes a '+ Add New Course' button and a 'Back' button. A table lists three courses with columns for ID, Courses, Subjects, Duration, Description, Fees (₹), and Action.

ID	Courses	Subjects	Duration	Description	Fees (₹)	Action
1	PG-DAC	Core Java, Spring Boot	6 Months	Post Graduate Diploma in Advanced Computing	90000	Edit Active
2	PG-DBDA	Big Data Hadoop	6 Months	Post Graduate Diploma in Big Data Analytics	95000	Edit Active
3	PG-DESD	Embedded C	6 Months	Post Graduate Diploma in Embedded Systems Design	88000	Edit Active

Subjects:

The screenshot shows the 'Subject List' page in the IMS Admin Panel. The left sidebar is identical to the previous page. The main content area is titled 'Subject List' and includes an 'Add New Subject' button and a 'Back' button. A table lists four subjects with columns for Subject ID, SUBJECT NAME, Courses, Assigned Teacher, Schedule, and Action.

Subject ID	SUBJECT NAME	Courses	Assigned Teacher	Schedule	Action
1	Core Java	PG-DAC	Prathamesh Pimpalkar	N/A	Edit Active
2	Spring Boot	PG-DAC	Prathamesh Pimpalkar	N/A	Edit Active
3	Big Data Hadoop	PG-DBDA	manish	N/A	Edit Active
4	Embedded C	PG-DESD	Prathamesh Pimpalkar	N/A	Edit Active

Notices:

ADMIN PANEL

Dashboard

Profile

Add User

Students

Teachers

Courses

Subjects

Notices

English

हिंदी

मराठी

Log Out

Manage Notices

Back

Add New Notice

Title

Enter notice title

Target Audience

All Users (General)

Date

mm/dd/yyyy

Description

Enter notice details...

Publish Notice

Recent Notices

CDAC Internal Exams

2026-01-31STUDENT

Active

Internal exams start from 15 March

Faculty Meeting

2026-01-31TEACHER

Active

All teachers must attend meeting on Friday

PG-DAC Lab Schedule

2026-01-31STUDENT

Active

PG-DAC lab sessions will start from next Monday.

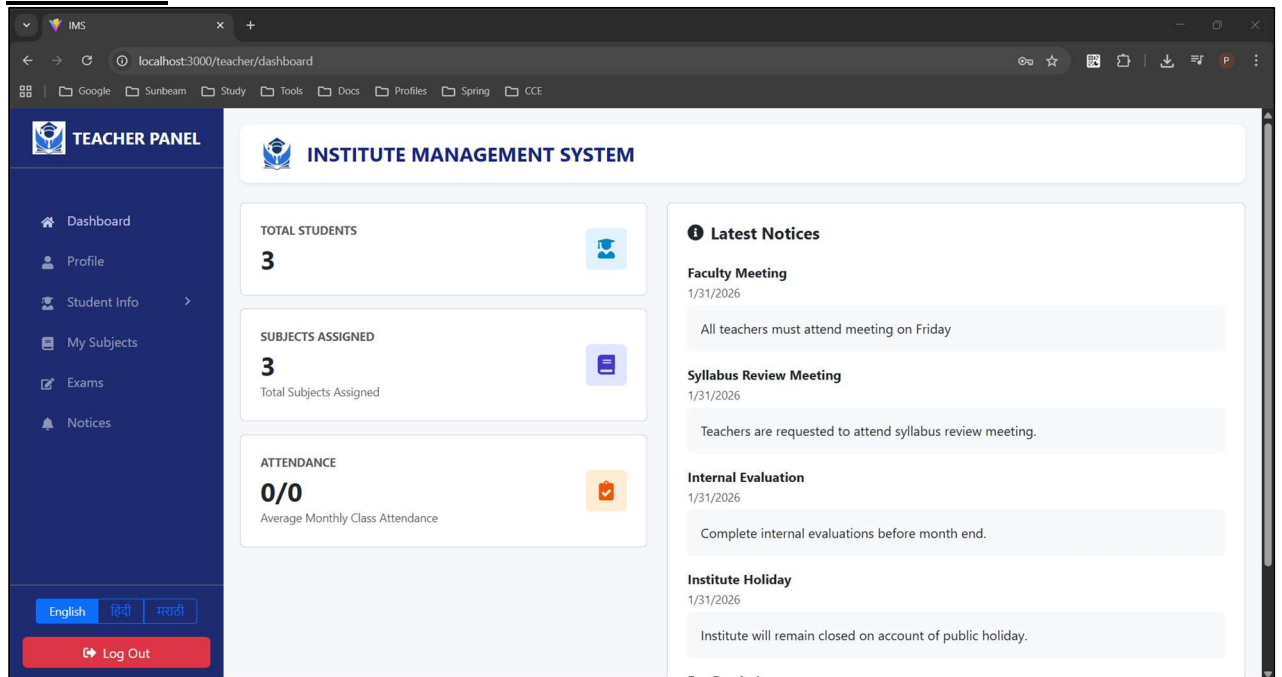
Assignment Submission

2026-01-31STUDENT

Active

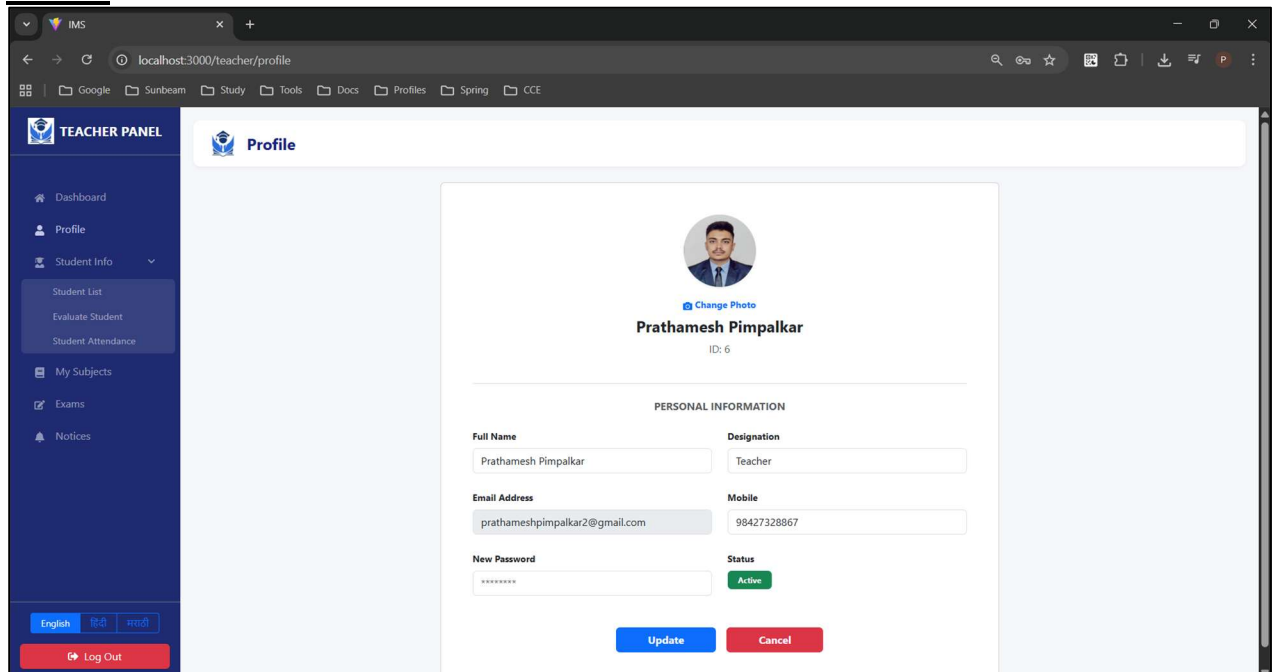
Teacher:

Dashboard:



The screenshot shows the Teacher Dashboard of the Institute Management System (IMS) in a web browser. The browser address bar shows the URL `localhost:3000/teacher/dashboard`. The page has a dark blue sidebar on the left with the "TEACHER PANEL" header and a menu containing: Dashboard, Profile, Student Info, My Subjects, Exams, and Notices. At the bottom of the sidebar are language buttons for English, हिंदी, and मराठी, and a red "Log Out" button. The main content area has a header "INSTITUTE MANAGEMENT SYSTEM" and displays three summary cards: "TOTAL STUDENTS" with a value of 3, "SUBJECTS ASSIGNED" with a value of 3 and a subtext "Total Subjects Assigned", and "ATTENDANCE" with a value of 0/0 and a subtext "Average Monthly Class Attendance". To the right of these cards is a "Latest Notices" section with four items: "Faculty Meeting" (1/31/2026) with the text "All teachers must attend meeting on Friday", "Syllabus Review Meeting" (1/31/2026) with the text "Teachers are requested to attend syllabus review meeting.", "Internal Evaluation" (1/31/2026) with the text "Complete internal evaluations before month end.", and "Institute Holiday" (1/31/2026) with the text "Institute will remain closed on account of public holiday."

Profile:



The screenshot shows the Teacher Profile page of the Institute Management System (IMS) in a web browser. The browser address bar shows the URL `localhost:3000/teacher/profile`. The page has the same dark blue sidebar as the dashboard, but the "Profile" menu item is selected. The main content area has a header "Profile" and displays a user profile for "Prathamesh Pimpalkar" with ID 6. Below the profile information is a "PERSONAL INFORMATION" form with the following fields: "Full Name" (Prathamesh Pimpalkar), "Designation" (Teacher), "Email Address" (prathameshpimpalkar2@gmail.com), "Mobile" (98427328867), and "New Password" (masked with dots). The "Status" is shown as "Active" in a green box. At the bottom of the form are "Update" and "Cancel" buttons.

StudentInfo:

Student List:

The screenshot shows the 'Student List' page in a web application. On the left is a 'TEACHER PANEL' sidebar with navigation links: Dashboard, Profile, Student Info (expanded), Student List (selected), Evaluate Student, Student Attendance, My Subjects, Exams, and Notices. At the bottom of the sidebar are language buttons (English, हिंदी, मराठी) and a Log Out button. The main content area is titled 'Student List' and features a search bar with a dropdown menu set to 'Search by Name' and a text input field labeled 'Type to search...'. A red arrow points to this search bar. Below the search bar is a table with the following data:

Roll No	Full Name	Email Address	Mobile	Gender	Courses	Join Date
STU1769842708526	arvin	arvin@gmail.com	654123123		PG-DAC	2026-01-31
STU1769842708526	arvin	arvin@gmail.com	654123123		PG-DBDA	2026-01-31
STU1769842708781	raman	raman@ims.com	984651794		PG-DAC	2026-01-31
STU1769842708781	raman	raman@ims.com	984651794		PG-DESD	2026-01-31
TEMP-1769842754397	Teerth Budhaliya	teerth@gmail.com	9842754397		PG-DBDA	2026-01-31
TEMP-1769842754397	Teerth Budhaliya	teerth@gmail.com	9842754397		PG-DAC	2026-01-31

This screenshot is identical to the one above, but the search bar dropdown menu is set to 'Search by Roll No'. A red arrow points to this dropdown menu. The table data remains the same as in the previous screenshot.

Roll No	Full Name	Email Address	Mobile	Gender	Courses	Join Date
STU1769842708526	arvin	arvin@gmail.com	654123123		PG-DAC	2026-01-31
STU1769842708526	arvin	arvin@gmail.com	654123123		PG-DBDA	2026-01-31
STU1769842708781	raman	raman@ims.com	984651794		PG-DAC	2026-01-31
STU1769842708781	raman	raman@ims.com	984651794		PG-DESD	2026-01-31
TEMP-1769842754397	Teerth Budhaliya	teerth@gmail.com	9842754397		PG-DBDA	2026-01-31
TEMP-1769842754397	Teerth Budhaliya	teerth@gmail.com	9842754397		PG-DAC	2026-01-31

Evaluate Student:

The screenshot shows the 'Evaluate Students' page in the Teacher Panel. The left sidebar contains navigation links: Dashboard, Profile, Student Info (with a dropdown), Student List, Evaluate Student (highlighted), Student Attendance, My Subjects, Exams, and Notices. At the bottom of the sidebar are language buttons (English, हिंदी, मराठी) and a Log Out button. The main content area has a header 'Evaluate Students' and two dropdown menus for 'Course' and 'Subject', both set to '-- Select --'. A 'Load Students' button is to the right. Below these is a table with columns: Roll No, Student Name, Marks, and Status. The table is currently empty, displaying 'No students loaded'.

Roll No	Student Name	Marks	Status
No students loaded			

Student Attendance:

The screenshot shows the 'Student Attendance' page in the Teacher Panel. The left sidebar is identical to the previous screenshot, with 'Student Attendance' highlighted. The main content area has a header 'Student Attendance' and a table with columns: Roll No, Student Name, Course, and Overall %. The table contains seven rows of student attendance data.

Roll No	Student Name	Course	Overall %
STU1769842708526	arvin	PG-DAC	100%
STU1769842708526	arvin	PG-DBDA	%
STU1769842708781	raman	PG-DAC	0%
STU1769842708781	raman	PG-DESD	%
TEMP-1769842754397	Teerth Budhaliya	PG-DBDA	100%
TEMP-1769842754397	Teerth Budhaliya	PG-DAC	%

My Subjects:

TEACHER PANEL

- Dashboard
- Profile
- Student Info
 - Student List
 - Evaluate Student
 - Student Attendance
- My Subjects**
- Exams
- Notices

English हिंदी मराठी

Log Out

Assigned Subjects

ID	Course Name	START DATE	Subject Name	Schedule
11	PG-DAC	2026-01-31	Core Java	N/A

My Attendance:

TEACHER PANEL

- Dashboard
- Profile
- Student Info
- my_attendance**
- My Subjects
- Exams
- Notices

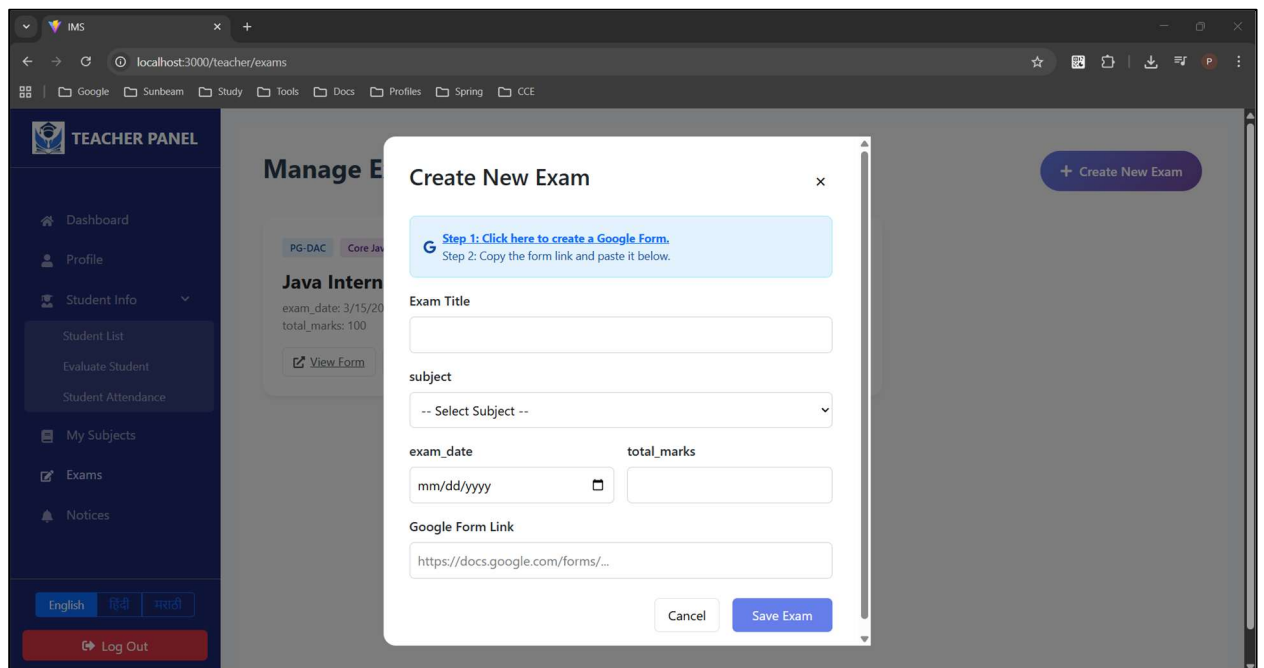
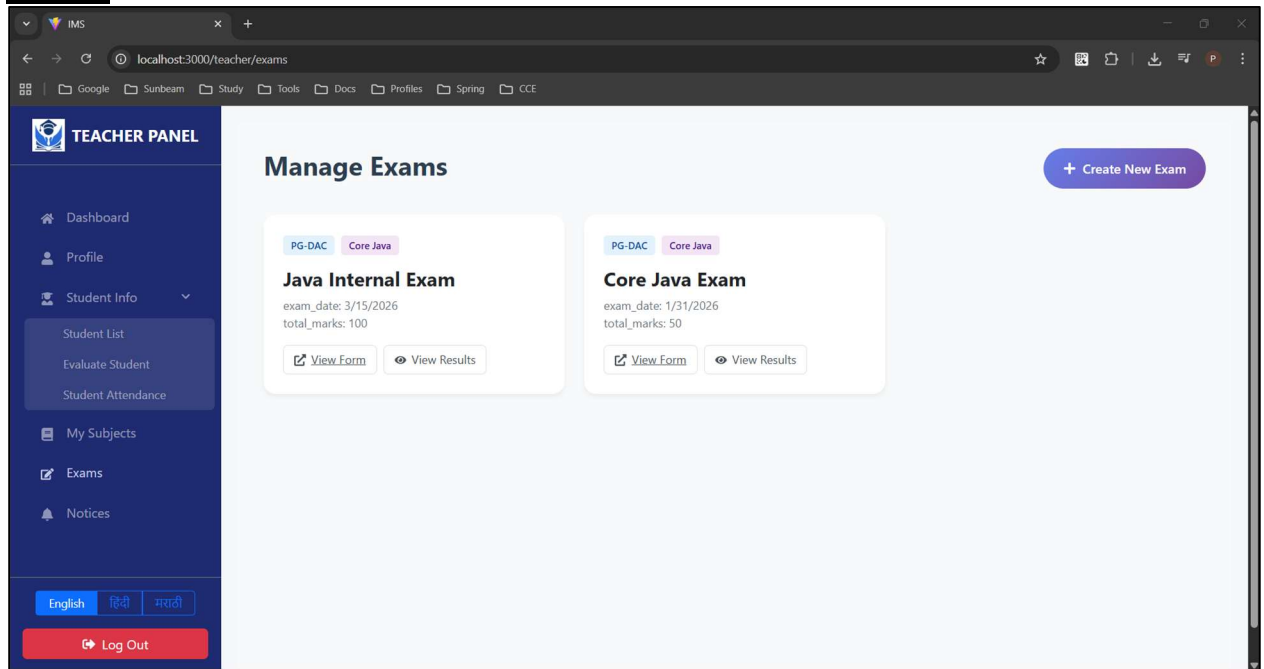
English हिंदी मराठी

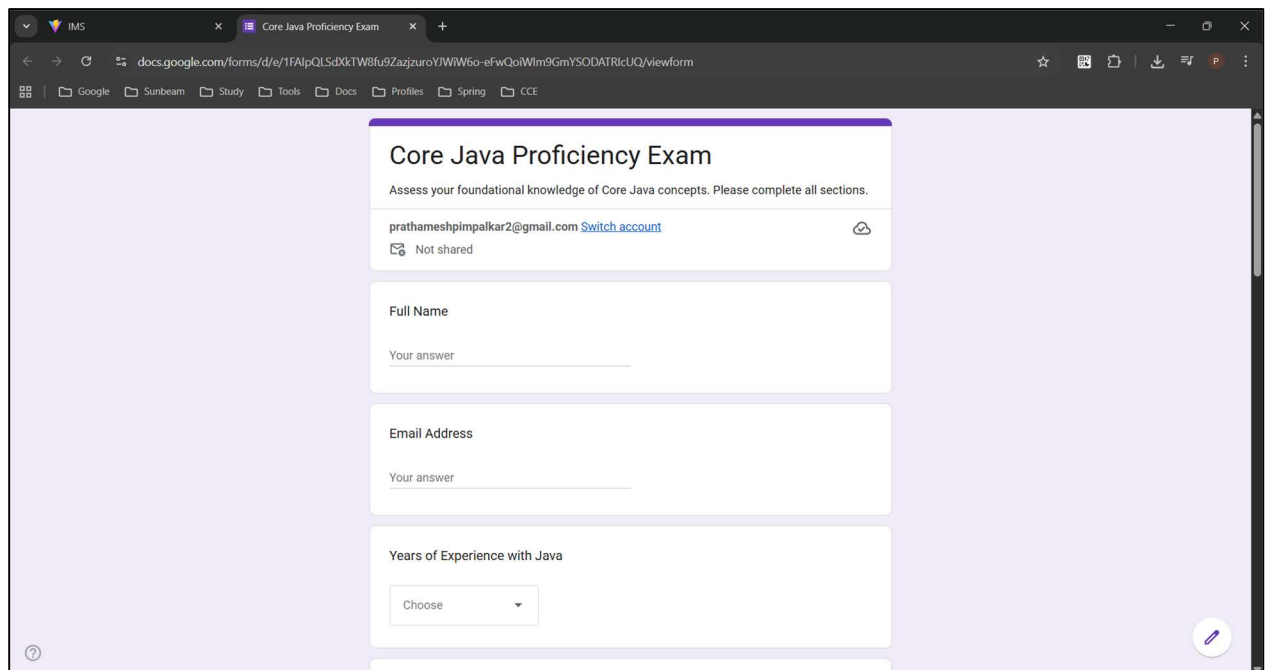
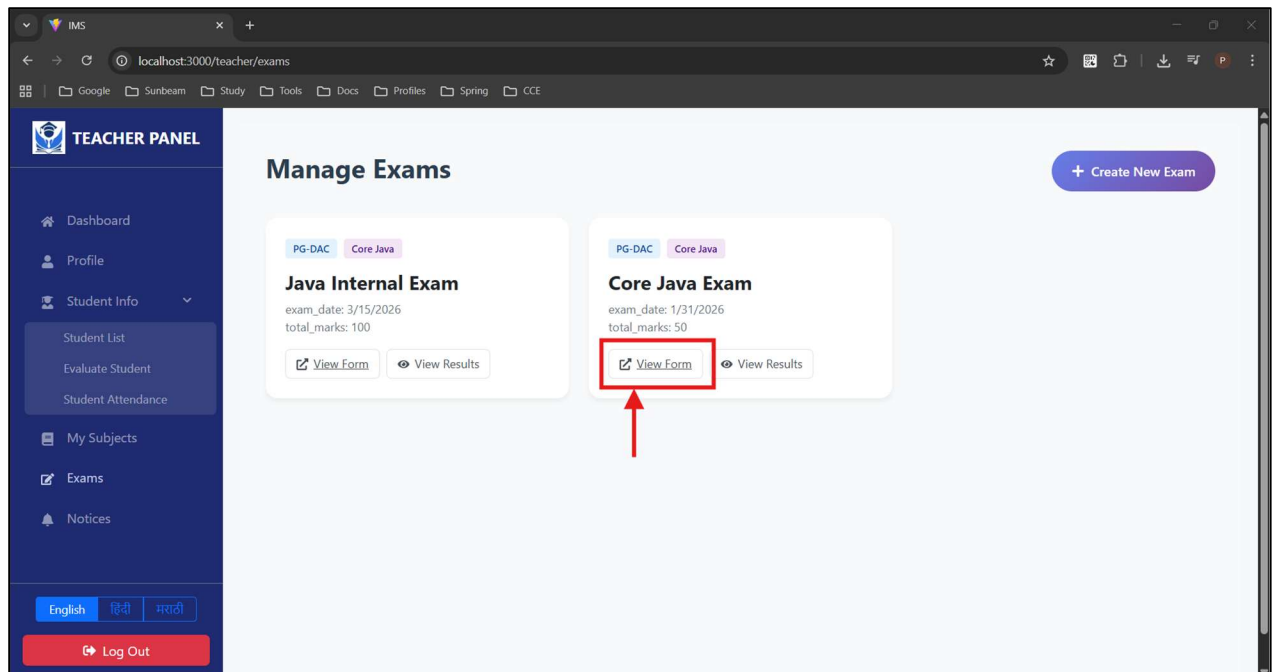
Log Out

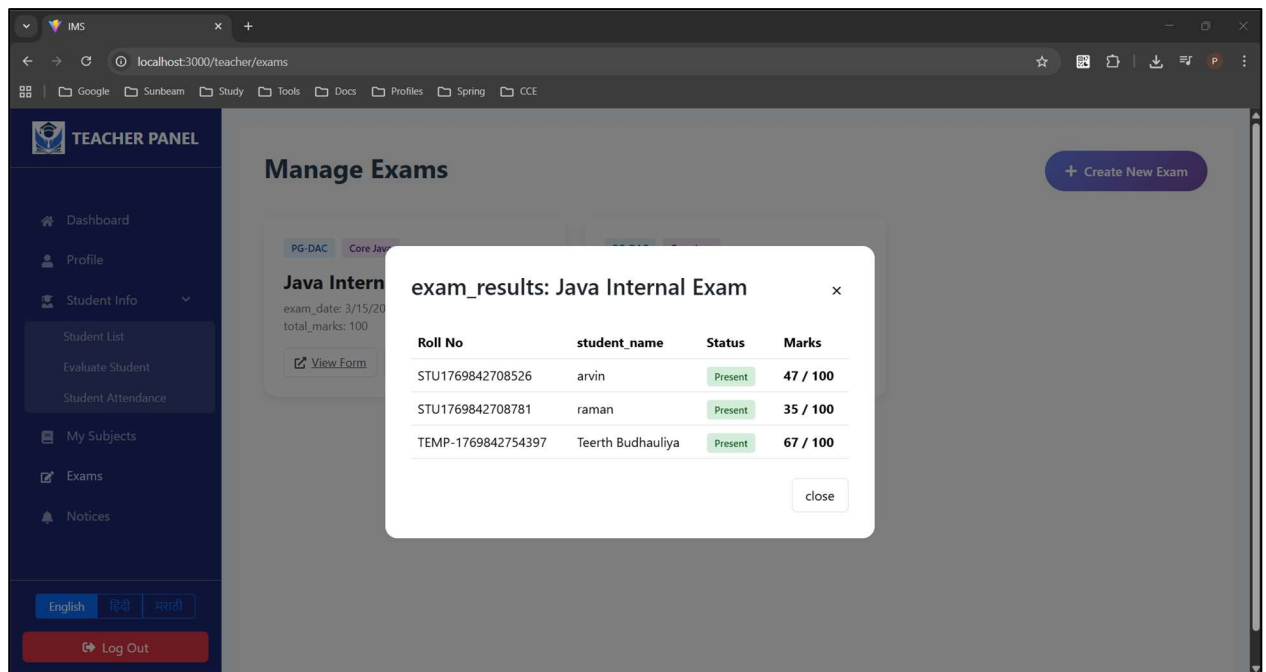
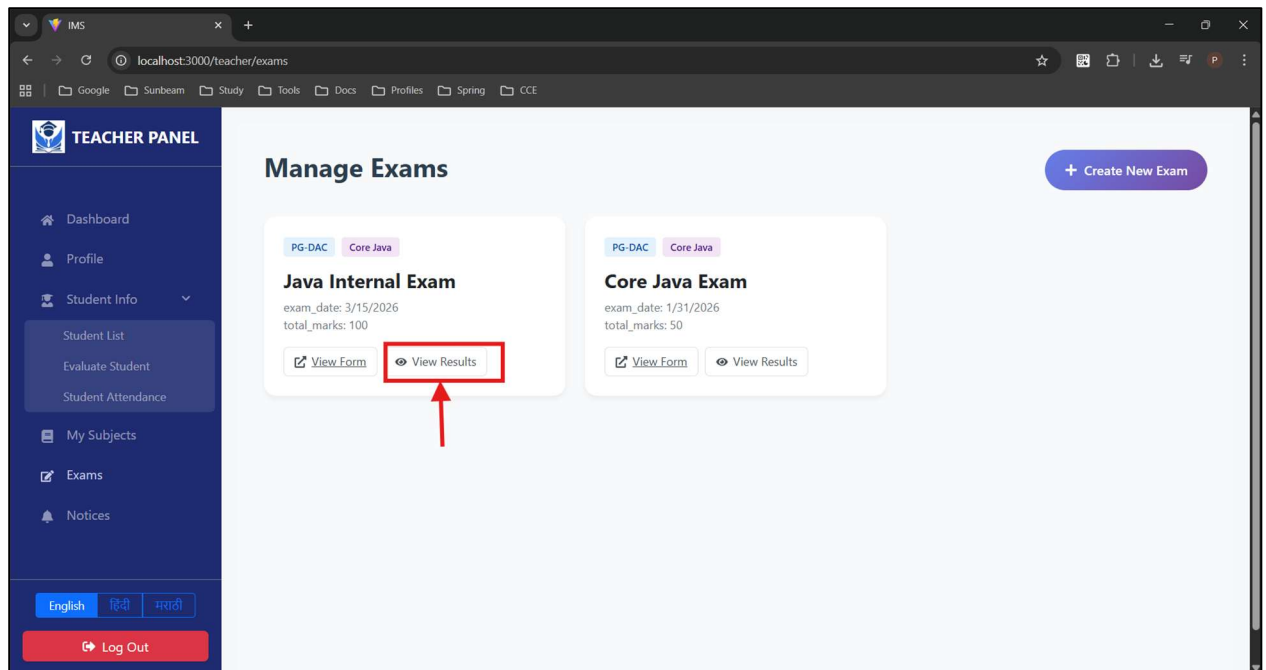
My Attendance Record

#	Date	Status
1	2026-02-05	LEAVE
2	2026-02-04	PRESENT
3	2026-02-03	ABSENT
4	2026-02-02	PRESENT
5	2026-02-01	PRESENT

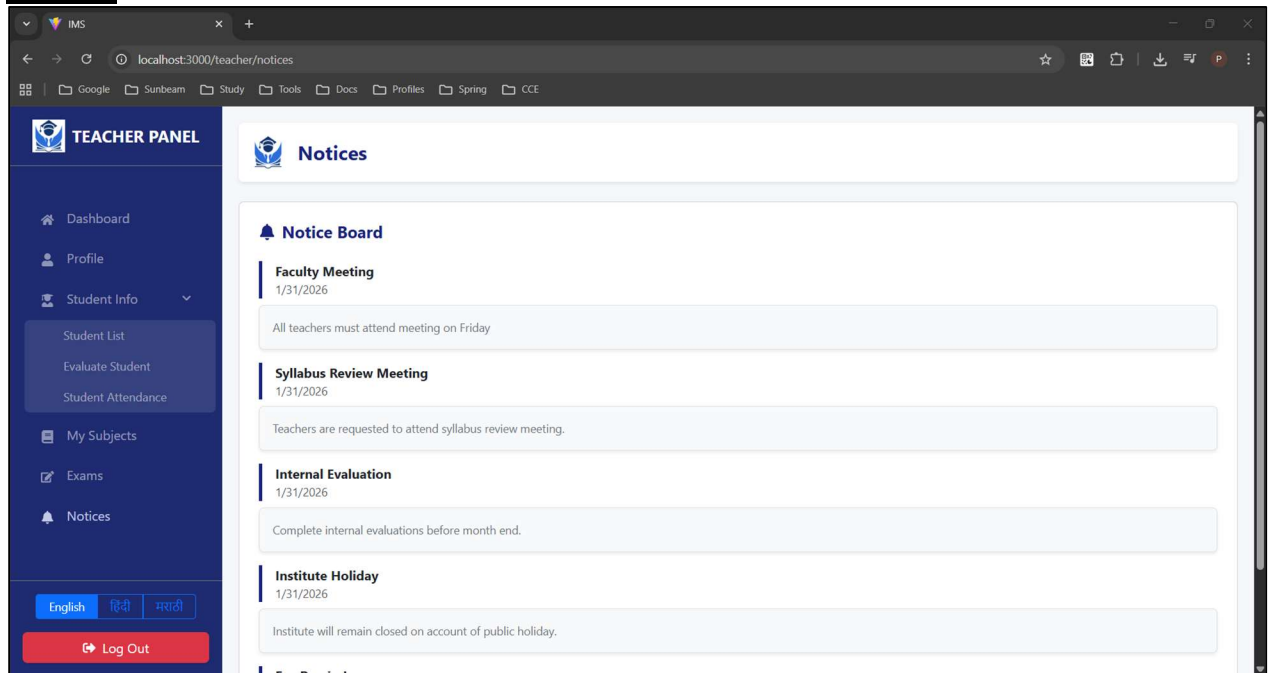
Exams:







Notices:



The screenshot shows a web application interface for a Teacher Panel. The browser address bar indicates the URL is localhost:3000/teacher/notices. The interface is divided into a left sidebar and a main content area.

TEACHER PANEL

- Dashboard
- Profile
- Student Info
 - Student List
 - Evaluate Student
 - Student Attendance
- My Subjects
- Exams
- Notices

Language options: English, हिंदी, मराठी. A Log Out button is at the bottom of the sidebar.

Notices

Notice Board

- Faculty Meeting**
1/31/2026
All teachers must attend meeting on Friday
- Syllabus Review Meeting**
1/31/2026
Teachers are requested to attend syllabus review meeting.
- Internal Evaluation**
1/31/2026
Complete internal evaluations before month end.
- Institute Holiday**
1/31/2026
Institute will remain closed on account of public holiday.

Student:

Dashboard:

The screenshot shows a web browser window with the URL `localhost:3000/student/dashboard`. The page features a dark blue sidebar on the left with the 'STUDENT PANEL' header and a list of navigation links: Dashboard, Profile, Subjects, Attendance, Marks, Exams, Notices, and Feedback. At the bottom of the sidebar are language selection buttons for 'English', 'हिंदी', and 'मराठी', and a red 'Log Out' button. The main content area is titled 'Dashboard' and includes a welcome message: 'Welcome back, here is your daily overview.' It contains three summary cards: 'ENROLLED COURSES' with a value of 2, 'TOTAL SUBJECTS' with a value of 2, and 'OVERALL ATTENDANCE' at 100.0%. To the right, a 'Latest Notices' section lists three items: 'CDAC Internal Exams' (2026-01-31), 'PG-DAC Lab Schedule' (2026-01-31), and 'Assignment Submission' (2026-01-31).

Profile:

The screenshot shows the 'Profile' page with the URL `localhost:3000/student/profile`. The sidebar is identical to the dashboard view. The main content area displays the user's profile for 'Teerth Budhaliya' with a roll number of 'TEMP-1769842754397'. It includes an 'Upload Photo' button. Below this, the 'STUDENT INFORMATION' section contains form fields for 'Full Name' (filled with 'Teerth Budhaliya'), 'Qualification' (filled with 'Not Provided'), 'Email Address' (filled with 'teerth@gmail.com'), and 'Mobile' (filled with '9842754397'). A 'Status' section shows 'Active' in a green box. An 'Update Profile' button is located at the bottom right of the profile section.

Subjects:

The screenshot shows a web application interface for a student panel. The left sidebar contains a 'STUDENT PANEL' header with a logo and a list of navigation items: Dashboard, Profile, Subjects, Attendance, Marks, Exams, Notices, and Feedback. At the bottom of the sidebar are language selection buttons for English, हिंदी, and ગુજરાતી, and a red 'Log Out' button. The main content area is titled 'Enrolled Subjects' and displays a table with the following data:

ID	Courses	START DATE	SUBJECT NAME	Schedule
1	PG-DBDA	2026-01-31	Big Data Hadoop	N/A
2	PG-DAC	2026-01-31	Core Java	N/A
3	PG-DAC	2026-01-31	Spring Boot	N/A

Attendance:

The screenshot shows the 'Attendance Tracker' section of the student panel. The left sidebar is identical to the previous screenshot. The main content area has a title 'Attendance Tracker' and includes filters for 'PG-DAC' and 'January'. Below the filters are four summary cards: 'TOTAL 0', 'PRESENT 0', 'ABSENT 0', and 'ATTENDANCE % 0.0%'. A table below these cards shows attendance records for the date '2026-01-31' for the subject 'Big Data Hadoop', with a status of 'PRESENT'.

Date	Subject	Status
2026-01-31	Big Data Hadoop	PRESENT

Marks:

IMS

localhost:3000/student/marks

GoogleSunbeamStudyToolsDocsProfilesSpringCCE

STUDENT PANEL

Dashboard

Profile

Subjects

Attendance

Marks

Exams

Notices

Feedback

Englishहिंदीमराठी

Log Out

Student Marks

ID	Course	Subject	Obtained	Total	Result
3	PG-DBDA	Big Data Hadoop	75	100	PASS
4	PG-DAC	Core Java	67	100	PASS

Overall %: 71.00%

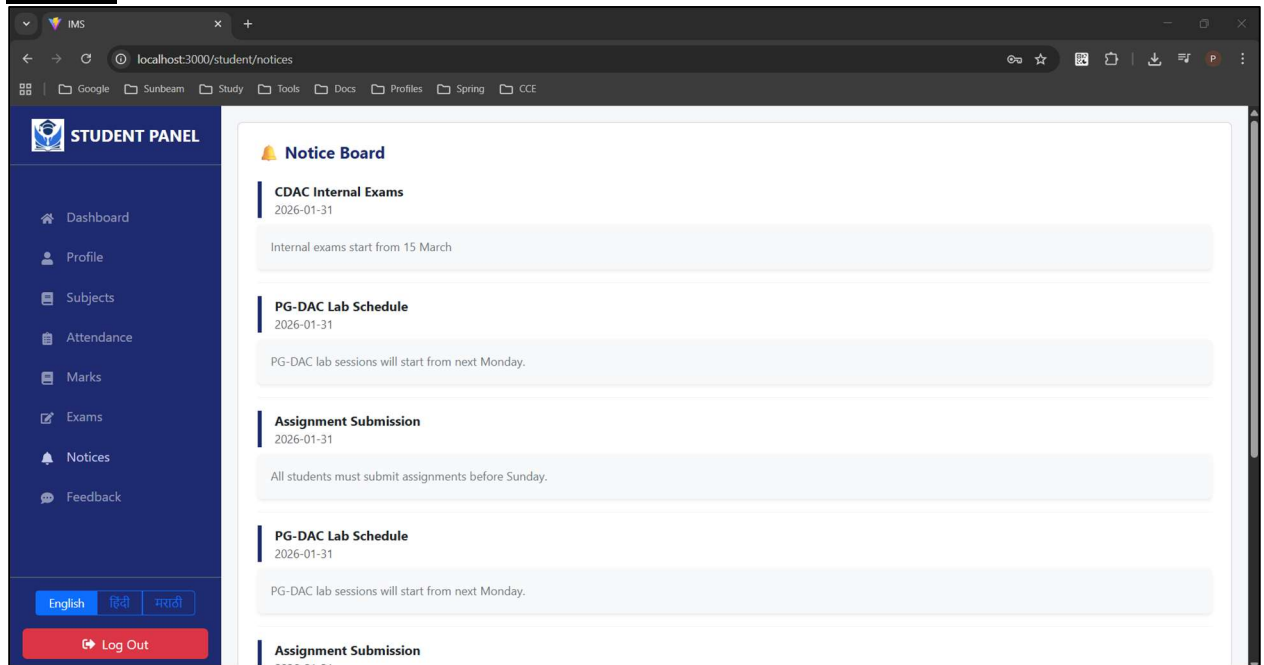
PASS

Exams:

The screenshot shows a web application interface for a student panel. On the left is a dark blue sidebar with the 'STUDENT PANEL' header and a list of navigation links: Dashboard, Profile, Subjects, Attendance, Marks, Exams, Notices, and Feedback. At the bottom of the sidebar are language selection buttons for English, हिंदी, and मराठी, and a red 'Log Out' button. The main content area is titled 'Available Exams' and displays three exam cards. The first card is for 'Java Internal Exam' (PG-DAC, Core Java) with a date of 3/15/2026 and marks of 67 / 100. The second card is for 'Core Java Exam' (PG-DAC, Core Java) with a date of 1/31/2026 and marks of 'Not Graded (50)'. This card is highlighted with a red border, and a red arrow points to its 'Take Exam' button. The third card is for 'Hadoop Test' (PG-DBDA, Big Data Hadoop) with a date of 3/20/2026 and marks of 75 / 100. Each card has a 'Take Exam' button with an external link icon.

The screenshot shows a Google Forms interface for the 'Core Java Proficiency Exam'. The form title is 'Core Java Proficiency Exam' with the instruction 'Assess your foundational knowledge of Core Java concepts. Please complete all sections.' The form is associated with the email 'prathameshpimpalkar2@gmail.com' and is marked as 'Not shared'. It contains three input sections: 'Full Name' with a text field labeled 'Your answer', 'Email Address' with a text field labeled 'Your answer', and 'Years of Experience with Java' with a dropdown menu labeled 'Choose'. A help icon is visible in the bottom left corner, and an edit icon is in the bottom right corner.

Notices:



The screenshot shows the 'STUDENT PANEL' interface for the 'Notices' section. The left sidebar contains links to Dashboard, Profile, Subjects, Attendance, Marks, Exams, Notices, and Feedback. The main content area is titled 'Notice Board' and displays a list of notices. Each notice includes a title, a date (2026-01-31), and a description. The notices are: 'CDAC Internal Exams' (Internal exams start from 15 March), 'PG-DAC Lab Schedule' (PG-DAC lab sessions will start from next Monday), 'Assignment Submission' (All students must submit assignments before Sunday), 'PG-DAC Lab Schedule' (PG-DAC lab sessions will start from next Monday), and 'Assignment Submission'.

STUDENT PANEL

- Dashboard
- Profile
- Subjects
- Attendance
- Marks
- Exams
- Notices
- Feedback

English हिंदी मराठी

Log Out

Notice Board

CDAC Internal Exams
2026-01-31
Internal exams start from 15 March

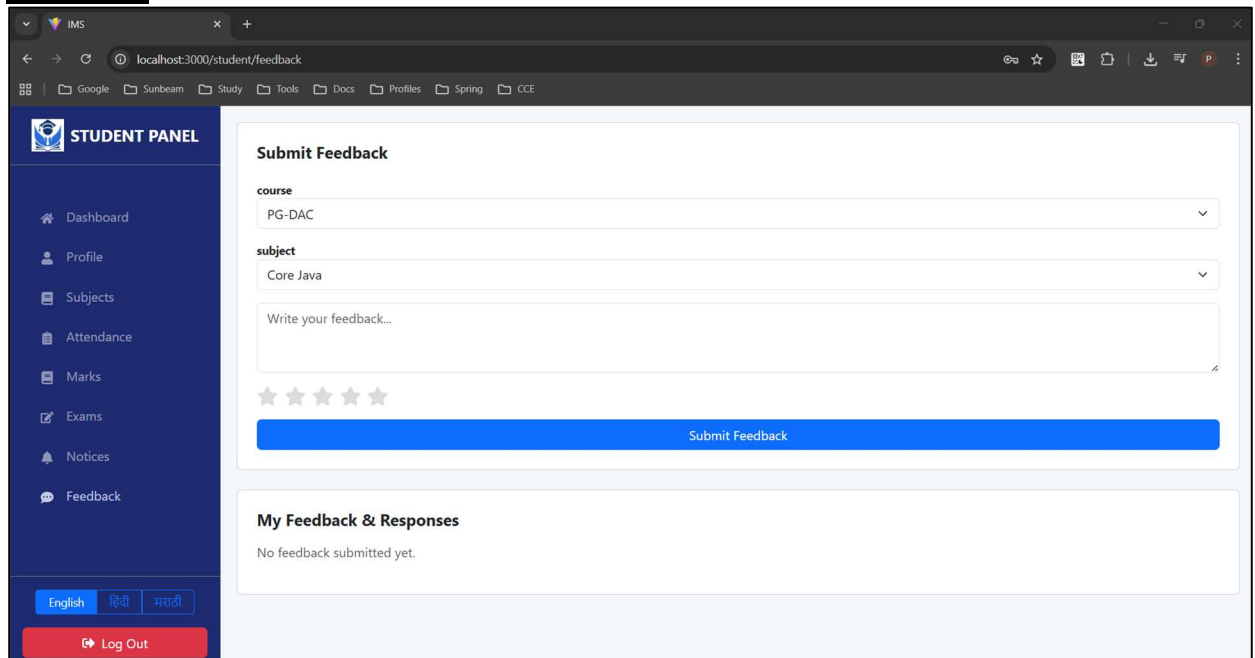
PG-DAC Lab Schedule
2026-01-31
PG-DAC lab sessions will start from next Monday.

Assignment Submission
2026-01-31
All students must submit assignments before Sunday.

PG-DAC Lab Schedule
2026-01-31
PG-DAC lab sessions will start from next Monday.

Assignment Submission

Feedback:



The screenshot shows the 'STUDENT PANEL' interface for the 'Feedback' section. The left sidebar contains links to Dashboard, Profile, Subjects, Attendance, Marks, Exams, Notices, and Feedback. The main content area is titled 'Submit Feedback' and includes a form with dropdown menus for 'course' (PG-DAC) and 'subject' (Core Java), a text area for 'Write your feedback...', a five-star rating system, and a 'Submit Feedback' button. Below the form is a section titled 'My Feedback & Responses' which currently shows 'No feedback submitted yet.'

STUDENT PANEL

- Dashboard
- Profile
- Subjects
- Attendance
- Marks
- Exams
- Notices
- Feedback

English हिंदी मराठी

Log Out

Submit Feedback

course
PG-DAC

subject
Core Java

Write your feedback...

★ ★ ★ ★ ★

Submit Feedback

My Feedback & Responses

No feedback submitted yet.

7.REFERENCES:

- 1) <http://www.google.com>
- 2) <http://students.sunbeamapps.org>
- 3) <https://docs.spring.io/spring-framework/reference/>
- 4) <https://docs.spring.io/spring-data/jpa/reference/index.html>
- 5) <http://www.webdevelopersjournal.com/>
- 6) <http://www.w3.org>
- 7) <http://www.wikipedia.org>
- 8) <https://telusko.com>